

Commercial Property Taxes, Firm Behavior, and Fiscal Externalities

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Abstract

Property taxation is widely viewed as a promising source of revenue for developing countries, yet evidence on its economic effects remains limited. This paper exploits a natural experiment in Kampala, Uganda, in which a property revaluation and mass enumeration increased tax liabilities within the city. Using administrative data from corporate and employee tax returns in a difference-in-differences framework, I document four main findings. First, the reform reduced profits, investment, employment, and survival probability, with effects that are highly heterogeneous across firm size and financial structure. Small firms experience declines in revenue and profits, whereas large firms experience revenue gains with no significant change in profits, consistent with greater market power. Financial flexibility also plays a critical role, insulating moderately leveraged firms from adverse effects. I also provide evidence that wage informality constitutes an important margin of adjustment, with a non-trivial share of formal job losses reflecting a shift to informal arrangements rather than outright job destruction. Second, the reform had progressive distributional consequences within firms, with workers at the top of the earnings distribution disproportionately affected. Third, the rental cost channel plays an important role in transmitting property tax burdens to firms. The reform increased business rental expenses by 6.5 percent, implying an estimated pass-through rate to formal tenant firms of 0.53. Finally, I quantify a vertical fiscal externality whereby each UGX 1 increase in local property tax revenue reduces central government income tax revenue by approximately UGX 0.26. The findings underscore a fundamental tension in decentralized fiscal systems: local revenue gains may impose costs on firms, workers, and the central government that are not fully internalized when tax instruments are evaluated in isolation.

JEL codes: H22, H24, H32, H71.

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I. INTRODUCTION

Property taxation is frequently highlighted as a promising revenue instrument in developing countries, with the potential to generate substantial revenue by improving compliance from currently low levels.¹ This has motivated a growing literature studying how administrative reforms and enforcement strategies can expand the property tax base and increase collections (e.g., [Manwaring and Regan 2025](#); [Flores et al. 2025](#); [Bergeron et al. 2024](#)). Yet, the focus on revenue has largely overlooked a central question: what are the real economic costs of property taxation?

From a theoretical perspective, property taxes are often viewed as relatively efficient, with the burden falling primarily on property owners and limited implications for real economic activity. This view, however, abstracts from two features that are particularly salient in many developing-country settings. First, it assumes a separation between property ownership and production. In practice, many firms occupy their own premises, so that property taxes effectively tax a productive input, raising the cost of using commercial space and potentially affecting firms' production decisions. Second, even in rental markets, factors such as mobility frictions and weak enforcement capacity may substantially alter tax incidence and generate meaningful pass-through to tenants. Consistent with this mechanism, recent evidence documents substantial pass-through of property taxes into rental costs (e.g., [Baker 2024](#); [Watson and Ziv 2024](#); [Löffler and Siegloch 2021](#); [Rolheiser 2019](#)). These cost increases may, in turn, induce firms to adjust along several margins, including employment, wages, and the use of owned versus leased space. The extent to which such adjustments arise, and whether they meaningfully alter real economic activity, remains an open empirical question.

These firm-level responses may, in turn, have broader fiscal implications that extend beyond the local government imposing the tax. If higher property taxes reduce firm profitability, induce exit, or contract formal employment and wages, they can erode the bases of centrally administered taxes such as the corporate income tax and the personal income tax. This creates a vertical fiscal externality that local governments do not internalize when setting tax policy.

This paper provides a comprehensive analysis of the economic effects of property taxation, spanning real production effects, incidence, and distribution. It also quantifies spillovers from local property taxes to central government revenues, highlighting the importance of vertical fiscal linkages for tax policy design.

¹In Uganda, it is estimated that only 10–11 percent of properties pay their annual property tax liability, a rate similar to that observed in many other cities in low-income countries ([Manwaring and Regan 2025](#)).

I exploit a natural experiment in Kampala, Uganda, where property tax liabilities increased differentially across locations within the city through a revaluation and mass property enumeration exercise that began in 2016/17. Prior to this exercise, the last valuation had been conducted in 2006, meaning that taxable property values had remained unchanged for a decade. During this period, numerous new properties had been established that were not captured in the old tax roll. The 2016/17 exercise led to the creation of a new property tax roll, which included additional properties and assigned higher rateable values to existing ones. As a result, incumbent properties faced higher property tax liabilities due to increased valuations, while properties not previously on the tax roll experienced an extensive margin change, although the marginal tax rate remained unchanged in both cases. Following the exercise, the total number of properties on the tax roll increased by 76 percent, while projected property tax revenue more than tripled.

This exercise was implemented in a staggered manner, starting with one division and gradually extending to others over time. For context, the capital city is divided into five administrative divisions. I exploit this geographical variation using a staggered difference-in-differences approach following [Callaway and Sant'Anna \(2021\)](#), with firms located outside the capital but within the Greater Kampala Metropolitan Area serving as never-treated units. This strategy allows me to identify the causal effects of increases in property tax liabilities on firm-level outcomes. The identifying assumption is that, in the absence of the reform, average outcomes for firms in treated and untreated locations would have evolved similarly, i.e., the parallel trends assumption. I provide evidence in support of this assumption using an event-study analysis. The analysis utilizes administrative tax data from corporate income and employee tax returns. These novel firm-level panel datasets span nine years (2013/14 - 2021/22) and were recently compiled by the Uganda Revenue Authority with support from UNU-WIDER.

First, I analyze how the reform affected business revenues, profits, and investment, focusing on a panel of active firms. Overall, I find no statistically significant effects on revenue. However, the reform reduced firms' profits by 2.4 percentage points relative to pre-reform revenues, while investment declined by 2.7 percentage points. The effects are very heterogeneous by firm size: revenues and profits decline for small firms, while large firms experience revenue gains, leaving their profits and investment unchanged. Although I cannot directly observe prices, this pattern is consistent with large firms exercising greater market power, enabling them to pass part of the tax burden onto consumers through higher output prices. This size-dependent heterogeneity in pass-through capacity has important implications for the distributional incidence of commercial property taxation: the burden falls disproportionately on smaller, less financially resilient firms,

while larger firms are relatively insulated.

To further understand firm dynamics, I examine the effects of the reform on firm survival. The firm survival outcome is constructed following the approach of [Saez et al. \(2019\)](#), restricting the sample to firms present in 2015/16 (the year prior to the reform) that reported non-zero turnover, regardless of whether they operated in other years. Firms are assigned a turnover value of zero in years in which they do not operate. The outcome variable is binary, equal to one if the firm operated in that year and zero otherwise. Consistent with [Saez et al. \(2019\)](#), the sample is reweighted using the nonparametric reweighting method developed by [DiNardo et al. \(1996\)](#) (DFL reweighting). The results indicate that the reform reduced firms' survival probability by 3.2 percent. Scaling this reduced-form effect by the aggregate increase in potential property tax liabilities implies that a \$1 million increase in aggregate property tax liabilities leads to approximately 25 additional firm exits citywide.

A potential threat to this approach of analyzing survival is that results could be compounded by evasion of corporate income tax (CIT). To address this, I implement a cross-tax validation strategy which classifies a case as exit if and only if the firm disappears from CIT and has no footprints in the employee tax returns. I further impose a stricter definition of exit, requiring no CIT return in year t and no record in employee tax returns in both years t and $t + 1$. Both of these restrictions show that the results are not driven by evasion responses. Moreover, heterogeneity analysis reveals that the negative survival effects are strongest among firms with no corporate income tax liability (i.e., loss-making firms). As an additional robustness check, I show that the results are similar when the sample is restricted to firms that were present and reported positive turnover in the first year of the sample, rather than to firms in the year immediately prior to the reform.

Using the same sample, I analyze employment effects, assigning employment a value of zero following firm exit and normalizing it relative to its pre-reform level to facilitate interpretation in percent changes. On average, the reform reduced employment by 3.7 percent among firms in treated areas relative to untreated areas. Scaling this reduced-form estimate by the aggregate increase in potential property tax liabilities implies that a \$1 million rise in aggregate liabilities is associated with approximately 135 fewer jobs citywide.² In a sample of firms active throughout the entire period, employment declines by 4 percent, indicating that the overall reduction in employment is not driven solely by firm exit but also by within-firm adjustments among continuing firms. As a robustness

²Because property taxes are levied at the property level and the data do not link tenant firms to landlords, I do not observe firm-specific property tax liabilities. As a result, I can not calculate firm-level employment elasticities with respect to property tax liabilities.

check, I re-estimate the employment effects in levels as well as log and inverse hyperbolic sine transformation.³ Both the event-study plots and average effects are consistent with those obtained using normalized outcomes. Consistent with the employment results, I find that the firms' wage bill declines by 3.8 percent following the reform.

The results further suggest that informality constitutes an important margin of adjustment for firms facing cost shocks. The reform led to an increase in informal wages of 5.1 percentage points from a pre-reform mean of 7.7 percent, suggesting that a non-trivial share of the observed decline in formal employment reflects transitions to informal arrangements rather than outright job destruction. This mechanism helps account for the large estimated employment effects, particularly in settings with weak enforcement capacity.

I then turn to understanding the distributional consequences of the reform at firm level by assessing which types of workers are most affected. The sample is restricted to active firms with at least five workers in the pre-reform period to ensure adequate support across the wage distribution. First, I examine how the share of workers, relative to pre-reform total employment, in fixed wage quartiles evolves following the reform. Second, I analyze how the firm-level wage distribution evolves around the reform in treated firms relative to untreated firms. The evidence indicates that workers at the top of the earnings distribution are disproportionately affected. In particular, following the reform, the share of workers in the top quartile declined by 2.8 percentage points relative to pre-reform total employment, while the share of workers in the 50th–75th percentile increased by 2.2 percentage points. The shares in other bins are unaffected. This pattern is consistent with skill downgrading within firms. Consistent with the above findings, I find that wages fall significantly at the top of the distribution while no significant effects are found at the bottom of the distribution. These results suggest that the reform reduced within-firm wage inequality and was therefore progressive at firm level.

Heterogeneity analysis reveals that pre-reform profitability and leverage are important moderators. Effects are stronger among less profitable firms and monotonically decreasing in profitability. For leverage, I find a U-shaped pattern in tax incidence across the leverage distribution: high-leverage firms are vulnerable through a financial distress channel, whereby higher property tax obligations tighten liquidity constraints associated with existing debt service obligations; very low-leverage firms are vulnerable through a financial exclusion channel, whereby limited access to credit markets constrains their ability to smooth the shock, forcing adjustment on real margins. Firms with moderate leverage retain sufficient unused debt capacity to absorb the shock, explaining their relative

³The log transformation is $\log(1+\text{employment})$ since employment can be zero in some years.

insulation. I provide direct evidence corroborating this mechanism: moderately leveraged firms experience a differential increase in debt following the reform, while firms at either tail of the leverage distribution do not. This asymmetry suggests that access to the borrowing margin is the key mechanism insulating moderate-leverage firms from the employment and survival consequences of the reform.

I further examine heterogeneity by occupancy status, estimating the effects on employment and survival separately for owner-occupiers and tenant firms. The reform reduced formal employment for both groups, though the effect is somewhat larger among tenants. Survival effects follow a similar pattern: while both groups exhibit negative point estimates, the effects are larger in magnitude and statistically significant only among tenant firms.

These larger effects among tenant firms suggest that the rental cost channel plays an important role in transmitting property tax burdens to firms. To quantify this channel, I estimate the effect of the reform on business rental expenses and calculate the implied pass-through rate. I consider an unbalanced panel of firms that reported paying rent in the year prior to the reform, estimating the effect of the reform on rent expenses scaled by their pre-reform values. The results reveal that the reform led to a 6.5 percent increase in business rental expenses. This reduced-form estimate implies that a UGX 1 increase in aggregate property tax revenue is associated with a UGX 0.53 increase in total rental expenses paid by CIT-registered tenant firms.⁴ Interpreted as a pass-through rate to the formal sector, this suggests that approximately 53 percent of the aggregate property tax increase is transmitted to formal tenant firms through higher rents, which likely represents a lower bound, since the denominator includes tax revenue from all commercial properties while the numerator captures rent increases among CIT-registered firms only.

This result contrasts with the neoclassical view, which predicts that the burden of property taxes should fall almost entirely on landlords. To rationalize this finding, I propose a framework that embeds a key institutional feature of property tax enforcement in developing countries: selective enforcement, whereby the probability of enforcement, and therefore compliance, is higher when a property is occupied and generating observable economic activity than when the unit is vacant. This pattern arises naturally in settings where tax administrations have limited capacity and prioritize enforcement on properties that are visibly in use. Under these conditions, property taxes raise the

⁴Because property taxes are levied at the property level and the data do not link tenant firms to landlords, I do not observe firm-specific property tax liabilities. As a result, it is not possible to construct a firm-level measure of tax pass-through. This pass-through rate is therefore estimated at the aggregate level for the population of CIT-registered firms. It should therefore not be interpreted as a structural, firm-level pass-through rate.

marginal cost of supplying space for rent, enabling landlords to shift a substantial share of the burden onto tenants.

Lastly, I quantify the vertical fiscal externality by studying how revenue from the corporate income tax and employee income tax changes around the reform. I find that total central government revenue collected from treated firms declines by an average of UGX 0.5 million annually relative to untreated firms, against a pre-reform mean of UGX 7.8 million, equivalent to a 6.4 percent decline. Using a back-of-the-envelope calculation, this effect is equivalent to UGX 0.26 of lost central government revenue for every UGX 1 increase in property tax revenues, suggesting that fiscal gains at the local level are substantially offset by revenue losses at the central level. This finding underscores the need for coordination across levels of government when designing sub-national tax policy, as the true fiscal yield of local tax instruments may be considerably lower than their face value once vertical externalities are taken into account.

Event-study analyses indicate that the results are not driven by differential pre-trends. The event-study plots indicate that pre-period coefficients are statistically indistinguishable from zero, suggesting the absence of differential pre-trends and that outcomes would have evolved similarly in treated and untreated areas in the absence of the reform. I also show that the results are not driven by COVID-19.

This paper contributes to three distinct strands of the literature. First, it contributes to the literature on how taxes affect business decisions. Existing studies have examined the effects of payroll taxes ([Saez et al. 2019](#); [Egebark and Kaunitz 2018](#)), corporate taxes ([Ljungqvist and Smolyansky 2014](#); [Kennedy et al. 2026](#)), business owner taxes ([Risch 2024](#)), and state and local tax policies in the United States ([Rohlin et al. 2014](#); [Thompson and Rohlin 2012](#); [Fajgelbaum et al. 2019](#); [Giroud and Rauh 2019](#)). Far less attention, however, has been devoted to understanding how property taxes affect real economic activity and its distributional consequences. To my knowledge, this paper provides the first comprehensive analysis of the economic effects of commercial property taxation, encompassing the real effects on firm activity, the incidence, and the distributional consequences within firms. It also provides novel evidence on the role of financial flexibility in moderating these effects.

Second, this paper contributes to the literature on spillover effects of tax enforcement. Existing work has focused primarily on horizontal spillovers, either within the same tax base through network effects, i.e., value added tax (e.g., [Pomeranz 2015](#); [Drago et al. 2020](#); [Boning et al. 2020](#)), or across taxes administered at the same level of government ([Lopez-Luzuriaga and Scartascini 2019](#)). I extend this literature along a vertical dimension, providing evidence that taxes administered at the sub-national level can erode the

revenue base of the central government, a fiscal externality that operates through firms' behavioral responses rather than through direct administrative linkages.

Third, this paper contributes to the long-standing literature on property tax incidence (see [England \(2016\)](#) and [Oates and Fischel \(2016\)](#) for reviews). While earlier work found low pass-through of property taxes to rent ([Carroll and Yinger 1994](#)), more recent studies have documented substantial pass-through to rental costs ([Watson and Ziv 2024](#); [Baker 2024](#)), particularly when the supply of housing is elastic ([Löffler and Siegloch 2021](#)). Prior research, however, has focused primarily on rental housing. This paper provides evidence on pass-through in the context of commercial properties.

The remainder of the paper is organized as follows. Section II describes the institutional context, including property taxation arrangements in Uganda and details of the revaluation and property enumeration reform. Section III describes the corporate income tax and employee tax return data used in the analysis and presents descriptive statistics. Section IV outlines the identification strategy. Section V presents the empirical results on the effects of the reform on various dimensions of business activity, including revenues, profits, investment, and firm survival, as well as labor market outcomes. Section VI examines the distributional consequences of the reform at firm level. Section VII presents heterogeneity along various dimensions. Section VIII presents empirical evidence on the effects of the reform on business rental expenses. Section IX quantifies the fiscal externality to central government revenues. Section X concludes.

II. CONTEXT

II.A. Property Taxation in Uganda

Property taxation in Uganda is governed by the Local Government (Rating) Act, 2005, which allows local jurisdictions to levy rates based on the rateable value of any property within their jurisdiction. The tax applies to all immovable property, including buildings and structures used for commercial purposes (e.g., rented residential houses, rented shops, factories, hotels). It also applies to owner-occupied properties used for business purposes but excludes vacant sites and owner-occupied residential properties.⁵ Local governments have the discretion to levy rates ranging from a minimum of UGX 2,000

⁵The full list of exempted properties includes: official residence of the president; official residences of cultural/traditional leaders; public worship places; cemeteries and crematoriums; charitable and educational institutions; outdoor sports and recreational facilities; local council-owned properties; property owned by privileged organizations under Diplomatic Privileges Act; International Treaty Tax-exempt organizations; and property belonging to any institution with which Government has contractual obligation not to levy fees and tax.

(approximately \$0.60) up to 12 percent of the rateable value, where the rateable value is defined as the assessed property value net of a 22 percent conservancy allowance. In Kampala Capital City Authority (KCCA), the property tax rate is set at 6 percent of the rateable value.

Currently, most cities and municipalities in Uganda, including Kampala, use a market-value approach in which property rates are applied to the estimated annual rental value or business income value (Bakibinga and Ngabirano 2019). The law requires each local government to prepare an initial valuation list and update it at least once every five years. In practice, however, these valuation lists are often not updated on schedule, largely due to a shortage of qualified valuation surveyors in both the private and public sectors (Bakibinga and Ngabirano 2019) and inadequate resources.

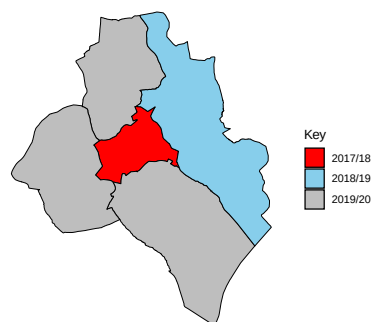
II.B. Property Revaluation, Enumeration, and Tax Collection in Kampala

To increase local revenue mobilization, the Kampala Capital City Authority (KCCA) initiated a citywide property revaluation and enumeration exercise beginning in the financial year 2016/17. Prior to this, the last valuation had been conducted in 2006, implying that the taxable values of properties remained unchanged for a decade. Over the same period, numerous newly developed properties were not captured in the existing tax roll. The number of properties on the tax roll increased by 76 percent, from 127,021 to 223,329, while the average property tax liability increased by 90 percent after the exercise. The revaluation and enumeration exercise was therefore expected to substantially expand tax revenue for KCCA, where property taxes had accounted for 35 percent of own-source revenue prior to the exercise. During the mass enumeration and survey exercise, KCCA collected detailed information on ownership, location (including GIS coordinates), and property attributes for valuation purposes. Property valuation was conducted primarily by professional valuers, although in some cases it relied on regression model predictions.

The exercise was implemented in a staggered manner, starting with one division and gradually extending to others over time. Figure 1 shows the rollout of the new tax roll across the five divisions of the capital city, following the revaluation and enumeration exercise. The different colors indicate the rollout phases of the exercise, as indicated in the legend.

The exercise resulted in a new property tax roll that both expanded coverage and increased the rateable values assigned to existing properties. As a result, incumbent properties faced higher property tax liabilities due to increased valuations, while properties not previously on the tax roll experienced an extensive-margin increase in tax liability, i.e., from zero to a positive tax liability, while the statutory tax rate remained unchanged in

Panel A: Kampala Capital City



Panel B: Greater Kampala Metropolitan Area

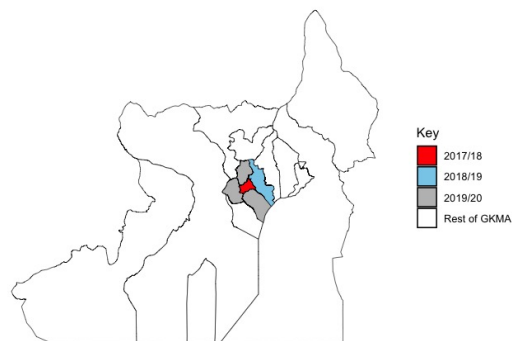


Figure 1: Kampala Capital City and Surrounding Areas

Notes: Panel A shows the five divisions of Kampala Capital City and the different phases of the rollout of the new property tax roll, as indicated in the legend. Panel B shows the Greater Kampala Metropolitan Area (GKMA), which includes counties in the neighborhood of the capital city. Firms in these locations serve as the never-treated units in the identification strategy, as these areas did not experience similar changes in property tax enforcement during the study period.

both cases. Figure 2 reports the number of properties and potential property tax revenue across the five divisions under both the old and new tax rolls. All divisions experienced an increase in the number of properties, with the number more than doubling in some divisions (Figure 2, Panel A). Similarly, potential revenue per property — a proxy for tax liability — increased across all divisions, with some divisions experiencing increases exceeding 180 percent (Figure 2, Panel B).

Property Tax Collection:— Following the valuation exercise, preliminary valuation lists are displayed in hard copy at the respective division offices, with additional dissemination through the press, radio announcements, and local leaders. Property owners are then granted a one-month period to file objections, after which the valuation court convenes to hear all appeals. Appeal decisions may result in either upward or downward revisions to tax liabilities. Final valuation rolls are then generated, and bills and demand notes are printed and delivered to property owners. Although the City Authority is not able to physically deliver demand notes to every property owner, the digital platform eCite sends SMS and email notifications to all billed property owners.⁶ The City Authority is also legally authorized to issue a demand notice to a tenant if the property owner is unavailable, and the tenant is legally permitted to deduct the tax payment from rental payments.

⁶Most of the information on the valuation and billing process was obtained from (Manwaring and Regan 2019).

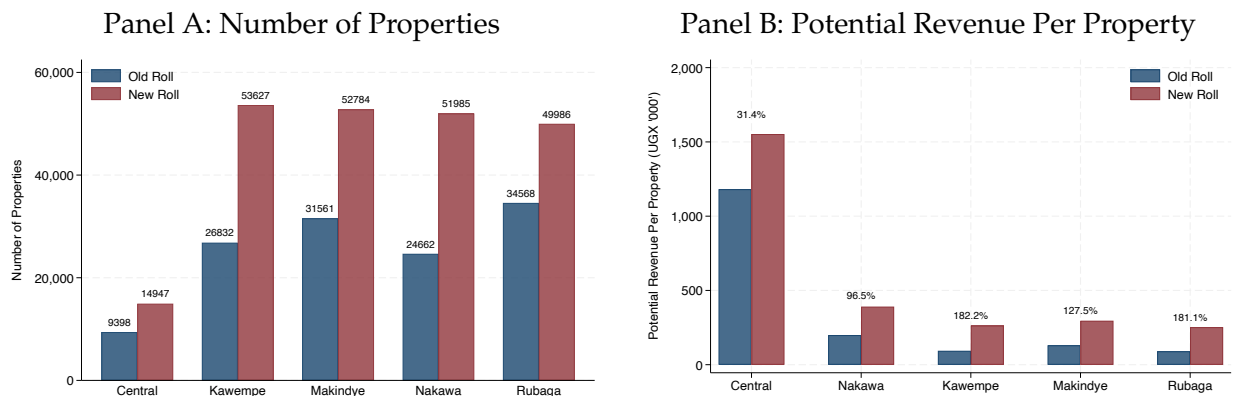


Figure 2: Number of Properties and Potential Property Tax Revenue

Notes: Panel A shows that number of properties in the old and new roll across the five divisions. Panel B shows the potential revenue per property before and after the reform. The data is extracted from [Ahabwe et al. \(2020\)](#) and [Bakibinga and Ngabirano \(2019\)](#).

Billing for all properties included on the final valuation roll occurs at the start of the financial year (July 1). Taxpayers have until December 31 to remit property tax payments without penalty. They may request payment plans, which are officially limited to a maximum of four equal installments. If payment is not made in time, KCCA may issue a demand notice requiring the taxpayer to settle the outstanding rate, along with any accrued interest, within two months of the notice being served. Interest begins accruing 30 days after the end of the financial year, at a rate of 2 percent per month, for as long as the liability remains unpaid. Payments can be made through commercial banks or via mobile money platforms.

III. DATA AND DESCRIPTIVE STATISTICS

III.A. Data Sources and Sample Construction

The paper uses administrative tax data from corporate income and employee tax returns as its primary datasets. The corporate income tax return (CIT) data comprise a firm-level panel covering the universe of firms registered for corporate income tax in Uganda, spanning the period 2013/14 – 2021/22. The employee tax return data cover the universe of individuals in formal employment over the same period. These are newly compiled datasets launched in 2022 by the Uganda Revenue Authority and UNU-WIDER.

The corporate income tax data contain annual balance sheets and profit and loss accounts, which provide detailed measures of firm activity, including revenues, profits, and

assets. All monetary values are recorded in Uganda shillings. To ensure comparability across years, I deflate them to constant 2016/17 values using the consumer price index (CPI). The data also include information on firms' sectors of operation and locations. Location information is central to treatment assignment, as discussed later in the paper. Because the corporate income tax data do not contain information on employment, I link them to the employee tax return data to construct a measure of formal employment by counting the number of workers filing under each firm. Since employee tax returns are filed monthly while corporate income tax returns are filed annually, I compute average employment over the twelve months of each year.

The full CIT dataset comprises *Sample Construction*:— The full CIT dataset comprises 404,527 firm-year observations. However, some firms report zero turnover throughout the entire sample period. Excluding these firms, the dataset is reduced to 299,683 firm-year observations, of which 81,162 (27 percent) form a balanced panel comprising 9,018 unique firms. The sample is further restricted to firms located in the Greater Kampala Metropolitan Area (GKMA) to ensure geographic comparability between treated and never-treated firms. Because the reform affects only firms in the capital city, firms in surrounding locations in the rest of the GKMA (Figure 1, Panel B) serve as the comparison group. This restriction yields 213,185 firm-year observations, representing 71 percent of firms that ever reported positive revenues during the sample period. Of these, 60,984 observations form a balanced panel comprising 6,776 unique firms, equivalent to 75 percent of the balanced-panel observations in the full dataset.

Additional sample restrictions vary by section or analysis. For the analysis of the effect on business rental expenses, the sample is restricted to firms that reported paying rent in the pre-reform period, as this outcome is normalized by its pre-reform value. For the baseline analyses of firm survival and employment effects, the sample is restricted to firms that were active and reported positive revenues.

III.B. Descriptive Statistics

Figure 3 plots the industry composition and size distribution across treated and comparison locations for firms that were active in the pre-reform year. The figure reveals substantial overlap along these dimensions. The firm size distribution is highly left-skewed: the majority of firms have annual revenues below UGX 150 million and fewer than five formal employees. However, the distribution is not systematically different for firms inside versus outside the city. Appendix Figure A.1 shows a similar overlap using density plots for revenues and assets. There are, however, some differences in industry composi-

tion across locations. For example, in retail and wholesale trade, the share of firms inside the city exceeds that outside the city by a considerable margin, while the reverse holds for the education sector. I flexibly account for differences in size and industry composition using fixed effects in the analysis. There is sufficient overlap in both dimensions to allow for comparisons of firms within the same industry and size bins across treated and comparison locations.

Table 1 reports pre-reform (2015/16) means and standard deviations for selected variables across different groups. The average firm in the universe of corporate income tax (CIT) filers has annual revenues of UGX 556 million and total assets of UGX 267 million, of which 70 percent are fixed assets. Average revenues for treated and comparison firms exceed those in the CIT universe, but are broadly comparable across the two groups. In contrast, firms in the comparison group hold more assets on average than treated firms, suggesting that firms in the city generate higher revenues per unit of assets. Similarly, average formal employment is higher among comparison firms than among treated firms. To account for these size-related differences, I include in the regressions time-invariant size bins defined by pre-reform employment. Nonetheless, robustness checks confirm that results are insensitive to the inclusion or exclusion of size fixed effects.

The average firm in the CIT universe pays annual rent of UGX 6.2 million and owns land and buildings worth UGX 72 million. Comparison firms have lower average rent expenses and are less likely to be renters (i.e., report positive rent expenses) compared to treated firms. In contrast, comparison firms have higher average values of land and buildings and are more likely to own property (report positive values for land and buildings). To assess the implications of these differences, I disaggregate all my main results by whether firms are tenants or owner-occupiers.

Lastly, I present statistics on variables capturing firms' financial positions. Firms with better financial situations may respond differently to cost increases associated with property tax liabilities. Table 1 shows that the share of loss-making firms is balanced at 23 percent across comparison and treated firms and slightly lower in the CIT universe. Similarly, average profit margins (EBIT/revenue) are similar across comparison and treated firms. Treated firms have higher liquidity (cash-to-assets) relative to comparison firms. While leverage is slightly higher in the CIT universe at 41 percent, it is balanced across treated and comparison firms. Overall, treated and comparison firms exhibit similar pre-reform financial characteristics.

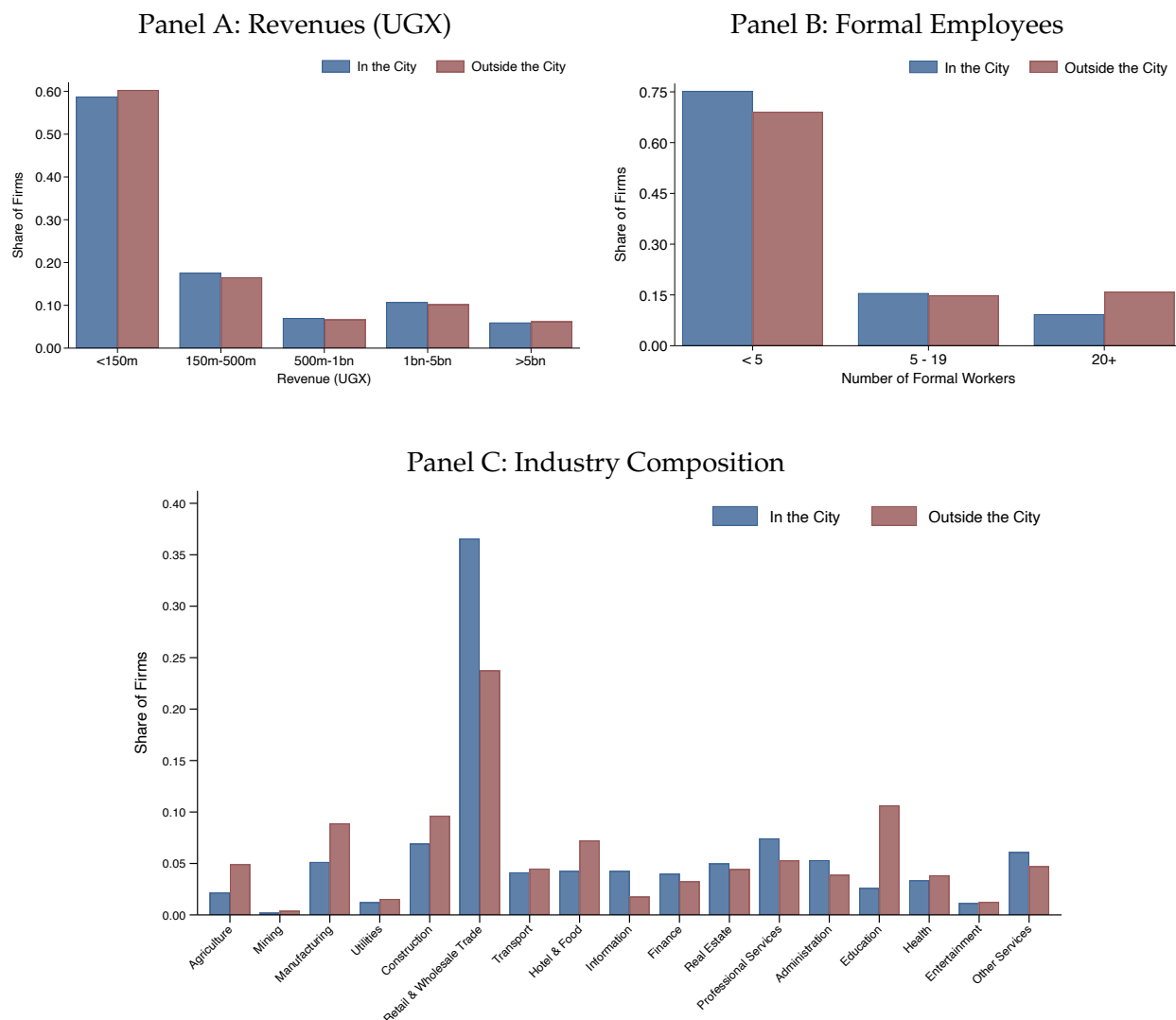


Figure 3: Firm Size Distributions and Industry Composition

Notes: This figure plots the industry and size mix of pre-reform active firms located inside (treated) and outside (comparison) the city. Each graph's bars sum to 1 within each location. Panels A and B show the distribution of revenues and employment, respectively. Panel C shows the industry composition of firms in the sample following the International Standard Industrial Classification of all Economic Activities (ISIC, Rev.4). Utilities combines electricity, gas, steam, and air conditioning supply with water supply, sewerage, waste management, and remediation activities. Administration combines administrative and support service activities with public administration & defense and compulsory social security. I exclude ISIC categories for activities of households as employers; undifferentiated goods- and services-producing activities of households for own use; and activities of extraterritorial organizations and bodies, as these account for negligible shares in the sample.

IV. IDENTIFICATION STRATEGY

I use a difference-in-differences design to estimate the causal impact of property tax enforcement on business rental expenses, business activity, and wages. Identification

Table 1: Summary Statistics

	(1) Universe	(2) Ever-Treated	(3) Comparison Group
Revenues	556.09 (1,232.30)	614.96 (1,294.14)	629.81 (1,328.00)
Total Assets	267.43 (595.70)	284.87 (620.41)	361.35 (696.65)
Fixed Assets	188.27 (461.14)	190.70 (468.84)	294.15 (568.66)
Formal Employees	5.16 (10.69)	5.15 (10.43)	7.45 (13.00)
Rent Expenses	6.15 (12.84)	8.39 (14.83)	4.43 (11.27)
Landing and Buildings	72.33 (209.02)	63.74 (202.33)	130.18 (270.43)
$\mathbb{I}\{\text{Rent Expense} > 0\}$	0.44 (0.50)	0.50 (0.50)	0.34 (0.47)
$\mathbb{I}\{\text{Landing and Buildings} > 0\}$	0.20 (0.40)	0.15 (0.36)	0.31 (0.46)
$\mathbb{I}\{\text{Loss Making}\}$	0.20 (0.40)	0.23 (0.42)	0.23 (0.42)
Profit Margins	0.15 (0.31)	0.13 (0.31)	0.14 (0.33)
Cash-to-Assets	0.17 (0.22)	0.17 (0.23)	0.13 (0.20)
Debt-to-Assets	0.41 (0.36)	0.35 (0.35)	0.36 (0.35)

Note: The table reports pre-reform (2015/16) means and standard deviations for selected variables across different groups. Standard deviations are reported in parentheses. Column 1 presents statistics for the CIT universe, while the remaining columns present statistics for different groups according to their treatment status. The ever-treated group consists of firms located in Kampala City, whereas the comparison group consists of firms from the rest of the GKMA. All monetary values are denominated in Uganda Shillings (UGX), constant 2016/17.

comes from comparing outcomes for firms located in Kampala with those for firms located in surrounding areas that were not exposed to changes in property tax liabilities during the sample period. The identifying assumption is that, absent the reform, average outcomes for firms in treated and untreated areas would have evolved similarly (i.e.,

the parallel trends assumption). I provide evidence in support of this assumption using event-study analysis.

A conventional approach would estimate the average and dynamic effects of the reform using Two-Way Fixed Effects (TWFE) difference-in-differences regressions, as in equations 1 and 2, respectively.

$$y_{it} = \alpha_g + \alpha_t + \beta D_{it} + \gamma X_{it} + \epsilon_{it}, \quad (1)$$

$$y_{it} = \alpha_g + \alpha_t + \sum_{\substack{e=-5 \\ e \neq -1}}^4 \beta_e D_{it}^e + \gamma X_{it} + \nu_{it}, \quad (2)$$

where y_{it} is the outcome variable. D_{it}^e is a time-varying indicator variable equal to one starting in the year firm i 's location received the reform and zero otherwise. α_g and α_t are group fixed effects and year fixed effects. X_{it} is a vector of controls.

However, recent literature shows that under staggered adoption, TWFE estimators are consistent only under homogeneous treatment effects. In the presence of heterogeneous effects across cohorts or over time, TWFE coefficients are generally biased and difficult to interpret (De Chaisemartin and d'Haultfoeuille 2020; Sun and Abraham 2021; Goodman-Bacon 2021; Callaway and Sant'Anna 2021; Borusyak et al. 2024). Bias in TWFE estimators arises from negative weighting induced by forbidden comparisons, whereby already-treated units are used as controls for later-treated units (Borusyak and Jaravel 2018). Heterogeneous effects are plausible in this setting for two reasons. First, firms' responses to increases in property tax liabilities may evolve over time as firms adjust along different margins. Second, changes in property tax liability vary across cohorts (Figure 2), implying that treatment effects may differ across cohorts.

To address these concerns, I estimate treatment effects using the doubly robust estimator proposed by Callaway and Sant'Anna (2021), with not-yet-treated units serving as the comparison group. This estimator remains consistent in the presence of cohort-specific treatment effects and accommodates conditional parallel trends, requiring only that treated and comparison units follow similar trajectories after conditioning on observed covariates. All results are robust to using never-treated firms as the comparison group instead, and to alternative estimators proposed by Sun and Abraham (2021) and Borusyak et al. (2024). Rather than estimating dynamic effects directly from the event-study regression in equation 2, the approach recovers group-time average treatment effects and aggregates them by exposure length, as in equation 3, thereby producing event-study estimates that are free from the bias induced by forbidden comparisons under TWFE.

$$\beta_{es}(e) = \sum_{g \in \mathcal{G}} \mathbb{I}\{g + e \leq \mathcal{T}\} \mathbb{P}(G = g \mid G + e \leq \mathcal{T}) ATT(g, g + e), \quad (3)$$

where, e denotes event time (the number of periods since initial treatment), G denotes the first period a unit is treated, and $ATT(g, t)$ is the average treatment effect for units who are members of a particular group g at a particular time period t , i.e., $ATT(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid X, G_g = 1]$. $\beta_{es}(e)$ is therefore the average treatment effect e periods after the treatment was adopted, averaging across all cohorts observed at event time e .

In estimation, I recognize that the rollout of the reform was not randomly assigned across groups. The City Authority initially targeted divisions with relatively high-value properties, namely Central Division, followed by Nakawa (Figure 2, Panel B). Moreover, the never-treated firms are located outside the capital city and may differ systematically from ever-treated firms. As shown in Table 1, firms in the capital differ on observables from those outside the city. Accordingly, I include baseline controls, size bins, sector, and accounting month in the regression specifications.

The key identifying assumption is that, in the absence of the property revaluation and mass enumeration exercise, outcomes of firms located within Kampala would have followed the same trajectory as those located outside the city. This parallel trends assumption does not require that the two groups of firms are identical in levels—treated and comparison firms may differ in size, sector composition, or baseline performance—but rather that average outcomes would have evolved along similar trajectories had the reform not occurred. I provide evidence consistent with this assumption using event-study estimates showing that the coefficients in the pre-reform periods are close to zero and statistically insignificant.

V. EFFECTS ON BUSINESS ACTIVITY

V.A. Revenues, Profits, and Investment

Higher property tax liabilities raise the cost of occupying commercial space, directly for owner-occupiers through higher tax bills and indirectly for tenants, to the extent that landlords pass the tax through to rents. These cost increases may propagate through broader dimensions of firm performance, compressing profit margins, reducing investment, and ultimately affecting revenues. To examine these effects, I estimate the reform's impact on log revenues, profits (scaled by pre-reform revenues), and investment rates, using a balanced panel of active firms.⁷

⁷Active firms are defined as those reporting non-zero turnover in at least one period.

Figure 4 presents the results. The absence of differential pre-reform trends across all three outcomes among firms in treated areas relative to those in untreated areas lends support to the parallel trends assumption underlying the identification strategy. Following the reform, I find no statistically significant average effects on revenues, while profits and investment decline markedly. Firms' profits fall by 2.4 percent relative to pre-reform revenues, and investment rates decline by 2.7 percentage points. The decline in profits is consistent with firms absorbing part of the burden associated with higher property taxes. Although the property tax is not purely a tax on businesses, this pattern aligns with evidence on local business tax incidence in settings with competitive product markets (Suárez Serrato and Zidar 2016). The decline in investment suggests that the cost shock not only compresses current profitability but also reduces capital accumulation, potentially generating persistent effects on firm productivity and growth. Appendix Figure A.4 corroborates these patterns, showing differential declines in value added, capital stock, and extensive-margin investment. Average effects are reported in Appendix Table A.1.⁸

These average effects, however, mask substantial heterogeneity by firm size. Appendix Figure A.3 shows that the profit and revenue declines are concentrated among small firms, while large firms experience revenue gains with profits and investment left largely unchanged. This pattern is consistent with larger firms exercising greater market power, enabling them to pass a share of the burden onto consumers through higher output prices. If product demand is sufficiently inelastic, such price adjustments can raise revenues even as costs increase. Small firms, by contrast, operate in more competitive environments or face more elastic demand, leaving them unable to adjust prices and therefore forced to absorb the increase in costs through lower margins. This heterogeneity in pass-through capacity by firm size has important implications for the distributional incidence of commercial property taxation: the burden falls disproportionately on smaller, less financially resilient firms, while larger firms are relatively insulated, and may even strengthen their competitive position relative to smaller rivals as a result.

V.B. Firm Survival

Beyond their effects on revenues, profits, and investment, property taxes may affect firms' continued operation in the local economy. A firm's decision to continue operating depends on whether expected future profits exceed the opportunity cost of remaining in business, a margin that a permanent increase in property tax liabilities can directly erode. Critically, unlike variable costs that adjust with output, property taxes are

⁸The investment rate is defined as the annual change in net fixed assets scaled by lagged fixed assets.

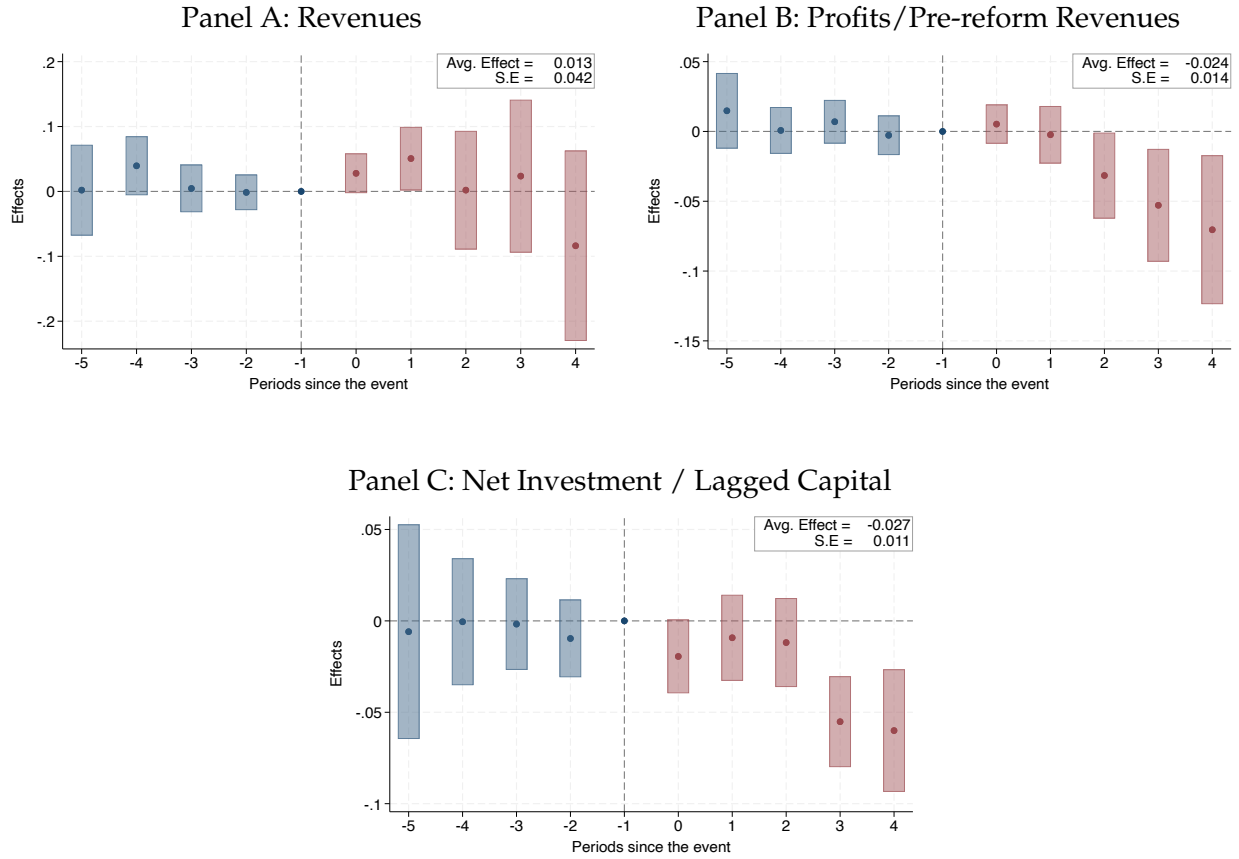


Figure 4: Effects of the Reform on Revenues, Profits, and Investment

Notes: The figure presents the dynamic effects of the reform on revenues, profits, and investment. Revenues are expressed in logarithms. Profits are measured as earnings before interest and taxes (EBIT), scaled by pre-reform revenues. The investment rate is defined as the annual change in net fixed assets, scaled by lagged fixed assets. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform size bins. The coefficients are estimated using a staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

largely fixed obligations that do not adjust with firm performance, making them particularly burdensome during periods of low revenue. The survival margin is therefore a key channel through which property taxation shapes the size and composition of the local firm economy. In this section, I investigate whether the reform affected the probability of firm survival.

Constructing the Survival Outcome:— Analyzing firm survival in firm-level panel data is challenging because firm exits inherently render the sample endogenous to the reform.

To address this, I restrict the cohort to firms present in 2015/16 (the year prior to the reform), with positive turnover, regardless of whether they operated in other years.⁹ I then construct a balanced panel by expanding each firm to all sample years and assigning zero values for turnover, sales, and related variables in the years the firm did not operate. The survival outcome is then defined as an indicator equal to 1 if a firm has positive turnover in year t and 0 otherwise. A similar approach was used by [Saez et al. \(2019\)](#) when analyzing whether payroll taxes affect firm survival in Sweden. By construction, the mean of the outcome variable is one in the reference year; it can be below one before 2016 because some firms had not yet entered, and below one after 2016 because some firms exit following the reform.

However, as highlighted by [Saez et al. \(2019\)](#), using this approach to estimate the effects on firm survival requires reweighting firms so that pre-reform distributions are aligned along key dimensions across different groups, since unweighted estimates can easily reflect compositional differences rather than the causal effects. For example, if younger firms, which tend to have lower baseline survival rates, are unevenly distributed across treated and untreated locations, unweighted estimates may simply capture age-related compositional differences rather than the effect of the tax reform itself. To address this, I reweight firms using the nonparametric reweighting method of [DiNardo et al. \(1996\)](#), following the implementation in [Yagan \(2015\)](#). Observations are first weighted by average pre-reform turnover, so that estimates reflect firms' economic scale and larger firms contribute proportionally more to the parameter estimates. I then reweight the sample across size-industry bins to ensure that treated and control groups are balanced along these dimensions in the pre-reform period, so that time-varying shocks to particular size or industry cells do not differentially influence the survival estimates.

Effects on Firm Survival:— Figure 5 shows the effects of the reform on firm survival using the reweighted sample constructed following the procedure described above. In the pre-reform period, there is no evidence of differential trends in survival probabilities between firms located in areas that experienced increases in property tax liabilities and those in areas that did not, as all pre-reform coefficients are statistically indistinguishable from zero. This finding supports the underlying identifying assumption that, absent the reform, survival probabilities for treated and untreated firms would have evolved similarly, supporting a causal interpretation of the post-reform estimates.

Following the reform, there is a differential decline in survival probabilities for firms

⁹As a robustness check, I repeat this procedure using the cohort of firms present in the first year of the sample (with positive turnover)

located in areas that experienced an increase in property tax liabilities relative to firms in locations that did not experience the reform, implying that the reform increased firm exit. The negative effects start to manifest after one year of the reform, grow in the second year before it stabilize at approximately 5 percentage points and persists up to the end of the sample. The corresponding average effects are presented in Table 2. Column 4 shows the preferred specification which includes the full set of controls. Comparing firms in locations exposed to increases in property tax liabilities with firms in other locations, the reform led to a 3.2 percent decline in survival probabilities with a standard error of 1.1 percent. Scaling this effect by the aggregate increase in potential property tax liabilities implies that a \$1 million increase in aggregate property tax liabilities leads to approximately 25 additional firm exits (see bottom row of Table 2).

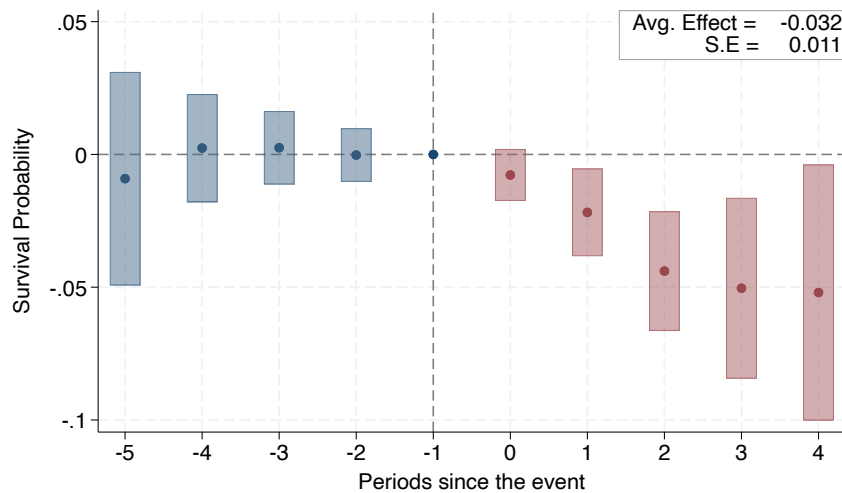


Figure 5: Effects of Property Tax Enforcement on Firm Survival

Notes: The figure presents the dynamic effects of the reform on firm survival. The outcome variable is an indicator equal to one if the firm is operating at time t , zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform size bins. The coefficients are estimated using a staggered Difference-in-Differences approach, following [Callaway and Sant'Anna \(2021\)](#), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

A key concern with this approach to measuring firm survival is that it may be confounded by evasion responses. Firms may respond to the higher costs associated with increased property tax liabilities by ceasing to file income tax returns, in which case this approach would misclassify them as exits. I implement several strategies to address this concern. First, I use a cross-tax validation strategy to identify firms' operational footprints. Using employee-level tax return data in combination with CIT data, I classify

Table 2: Effects of Property Tax Enforcement on Firm Survival

	(1)	(2)	(3)	(4)
ATE	-0.036*** (0.011)	-0.029*** (0.011)	-0.029** (0.011)	-0.032*** (0.011)
Sector of Operation	No	Yes	Yes	Yes
Pre-reform Size	No	No	Yes	Yes
Other Controls	No	No	No	Yes
Observations	130864	123864	123864	123864
Exits per \$1 Million	28.35	22.90	22.71	25.34

Note: The table presents average effects of the reform on firm survival probabilities. The outcome variable is an indicator equal to one if the firm is operating at time t , zero otherwise. Because the mean of the dependent variable is one in the pre-reform year by construction, the coefficient estimates can be interpreted as a percent change in survival probability. Each column reports estimates from a different specification with an alternative set of controls. Under “Other Controls,” I include dummies for accounting month, whether the firm reported owning land and buildings, and whether the firm filed returns on time, all measured in the pre-reform period. The bottom row reports the citywide reform-induced firm exits per additional \$1 million in aggregate property tax liability, calculated as $\frac{\hat{\beta} \cdot N}{\Delta T^{\text{City}}}$, where N denotes the total number of active firms in the city prior to the reform and ΔT^{City} is the reform-induced change in citywide property tax liability. The coefficients are estimated using a staggered Difference-in-Differences approach following [Callaway and Sant’Anna \(2021\)](#), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level.

evasion as cases where the firm disappears from the CIT but still has employees associated with its tax identification. Accordingly, I classify a case as exit if the firm disappears from the CIT and has no footprints in the employee tax returns, i.e., both tax records go silent. Appendix Figure A.5 shows that my estimates are unlikely to be driven by an evasion response. A likely criticism of this validation approach is that firms may collude with workers to evade both taxes. However, the timing mismatch between the two tax returns makes such collusion practically difficult. Employee tax returns are filed monthly, whereas corporate income tax (CIT) is filed annually. Since I classify a case as evasion even if the firm’s identifier appears in only one month, for collusion to occur, the firm would have to decide to evade at the start of the income year, before the realization of revenues. Moreover, workers would need to be compensated for agreeing to collude, as doing so would entail forfeiting other benefits such as social security. This creates an additional friction limiting firms’ ability to engage in coordinated evasion. To my knowledge, this paper is the first to cleanly separate survival effects from evasion response using cross-tax referencing.

Other Robustness Checks:— I conduct several robustness checks to assess the sensitivity of the survival effects, with results reported in the Appendix. First, I redefine exit as

the absence of a corporate income tax (CIT) return for at least two consecutive years, thereby excluding firms that temporarily disappear for one year before subsequently re-entering. Second, I impose a stricter definition of exit that requires no CIT return in year t and no record in employee tax returns in both years t and $t + 1$. As shown in Figure A.6, the estimated effects remain stable under both definitions. Third, I reconstruct the survival outcome following the same procedure as before but restrict the sample to the cohort of firms that were present and reported positive turnover in the first year of the sample. Figure A.7 shows that the results are consistent with those obtained using the baseline sample. As a final robustness check, I assess whether the findings are sensitive to the choice of comparison group in the Callaway and Sant’Anna (2021) difference-in-differences design. Figure A.8 shows that the results remain robust when using never-treated firms as the comparison group rather than not-yet-treated firms.

Net Effects:— The above analysis focuses only on firm exits and therefore does not capture net effects operating through both entry and exit. A reform that raises operating costs may simultaneously accelerate exit among incumbents and deter entry by potential entrants, both of which would reduce the stock of active firms. To examine these net effects, I construct a balanced panel at the county–industry level, using four-digit industry classifications, and analyze how the number of firms evolves before and after the reform across locations that experienced an increase in property tax liabilities relative to those that did not. I define the outcome variable as the log number of firms in each county–industry cell, which captures both incumbent exit and reduced entry. The results, presented in Figure A.9, show a differential decline in the number of firms operating in treated locations relative to untreated areas following the reform. This finding complements the survival analysis by confirming that the reform reduced the total stock of active firms, not merely the survival rates of incumbents. The decline in firm counts therefore reflects a combination of accelerated exit and reduced entry, both of which are consistent with higher operating costs reducing the returns to formal business activity in treated locations.

V.C. Labor Market Outcomes

Effects on Formal Employment:— The preceding results on firm survival have direct implications for employment. Firm exits mechanically reduce employment, but even among surviving firms, higher operating costs may induce workforce adjustments independently of the exit margin. I examine each of these margins in turn. I begin by analyzing the total employment impact using the broader sample from Section V.B, which includes exiting firms, assigning employment as zero following exit. This approach cap-

tures the full employment consequences of the reform, encompassing both job losses driven by firm closure and workforce adjustments among continuing firms. I then examine whether surviving firms adjust employment independently of the exit margin, focusing on a balanced panel of firms that remain in operation throughout the sample period — that is, firms with positive turnover in every year.

Because I observe only formal employment, which can be zero for some firms even prior to exit, and because employment is coded as zero after firm exit, I do not transform the outcome using logarithms in the main specification, as this would complicate the interpretation of coefficients as percentage effects, as discussed by [Chen and Roth \(2024\)](#). Instead, I normalize employment relative to its pre-reform level (2015/16), setting the normalized outcome to zero when employment is zero in both year t and the pre-reform year. I then winsorize the normalized outcome at the 95th percentile. Normalizing outcomes by pre-reform values is common in the literature (e.g., [Saez et al. 2019](#); [Harju et al. 2022](#)) and facilitates an economically meaningful interpretation of the regression coefficients in percent changes or percentage points. As a robustness check, I re-estimate the employment effects using alternative transformations of the outcome variable: $\log(1+\text{employment})$, the inverse hyperbolic sine transformation, and levels. Appendix Figure [A.10](#) shows that the event-study estimates and average treatment effects are consistent across all three transformations, with only minor differences in the magnitude of the coefficients, confirming that the baseline results are not sensitive to the choice of outcome transformation.

Figure [6](#) presents the event-study estimates and the corresponding 95% confidence intervals for the impact of the reform on formal employment, estimated following [Callaway and Sant'Anna \(2021\)](#). Prior to the reform, formal employment among firms in treated locations follows trends similar to those among firms in untreated locations. Following the reform, employment declines differentially among firms in treated areas relative to those in untreated areas. The negative effect emerges in the reform year and continues to grow through the end of the sample.

The average effects of the reform on employment are reported in Table [3](#). On average, employment in firms located in treated areas declined by 3.7 percent relative to those in untreated areas (Column 3). Appendix Figure [A.11](#) shows that employment declines even when the sample is restricted to firms active throughout the entire sample period, indicating that the aggregate employment effect is not solely driven by firm exit but also reflects deliberate workforce reductions among surviving firms. These within-firm adjustments among surviving firms signal that even firms that weather the reform do so partly by contracting their workforce in response to the higher cost burden. Together, these two margins suggest that the employment consequences of property taxation extend beyond

the extensive margin of firm survival documented in the previous section.

To assess the economic magnitude of these effects, I translate the estimated percent decline in employment into an aggregate citywide measure. Because property taxes are levied at the property level and the data do not link tenant firms to landlords, I do not observe firm-specific property tax liabilities. Moreover, multiple firms may operate within the same property. I therefore cannot estimate firm-level employment elasticities with respect to property tax liabilities. Instead, I aggregate the reduced form employment effects across firms and scale them by the aggregate increase in potential property tax liabilities induced by the reform. Following this approach, the reduced form estimate implies that a \$1 million increase in aggregate property tax liabilities is associated with approximately 135 fewer formal jobs citywide (see bottom row, Table 3, column 3).

While not directly comparable owing to differences in the tax instruments studied, the populations affected, and the institutional settings, the negative employment effects documented here are broadly consistent with estimates from related tax literature, including studies of payroll taxes in Sweden (Saez et al. 2019; Egebark and Kaunitz 2018) and corporate taxes in the United States (Ljungqvist and Smolyansky 2014; Giroud and Rauh 2019).

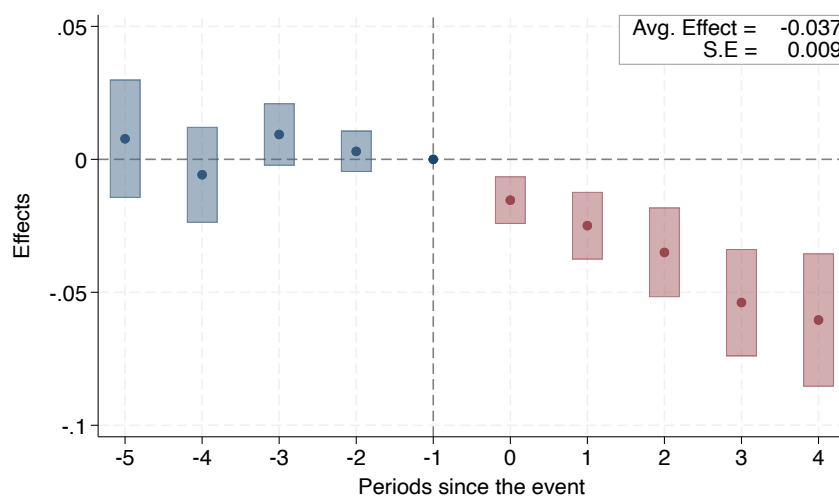


Figure 6: Effects of the Reform on Formal Employment

Notes: The figure presents the effects of the reform on employment. The dependent variable is normalized by its pre-reform value, implying that the coefficients can be interpreted as percentage changes. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform size. Standard errors are clustered at the firm level.

A potential concern with my analysis of employment effects is that workers displaced from treated firms may find employment at comparison firms, inflating the estimated

Table 3: Effects on Formal Employment

	(1)	(2)	(3)
ATE	-0.041*** (0.009)	-0.038*** (0.009)	-0.037*** (0.009)
Industry-Size	Y	Y	Y
Accounting Month	N	Y	Y
Other Controls	N	N	Y
Pre-reform Mean	6.19	6.19	6.19
Observations	123921	118584	118584
Jobs Lost per \$1 Million	150	139	135

Note: The table presents the average effects of the reform on the employment, normalized by its pre-reform value. Each column reports estimates from a different specification with an alternative set of controls. Under “Other Controls,” I include dummies for pre-reform profitability, occupancy status (whether the firm reported owning land and buildings or renting), and whether the firm filed returns on time prior to the reform. The bottom row reports the citywide reform-induced job loss per additional \$1 million in aggregate commercial property tax liability, calculated as $\frac{\hat{\beta} \cdot \bar{E} \cdot N}{\Delta T^{\text{City}}}$, where $\hat{\beta}$ is the reduced form coefficient, $\bar{E}$ denotes average pre-reform employment, N denotes the total number of active firms in the city prior to the reform and ΔT^{City} is the reform-induced change in citywide commercial property tax liability.

treatment effect through labor reallocation rather than genuine job destruction. I address this through a placebo analysis, restricting the sample to firms located outside Kampala city and comparing those within the greater Kampala metropolitan area (GKMA), which are geographically proximate to treated firms and therefore most likely to absorb displaced workers, against firms located outside the GKMA entirely.¹⁰ If reallocation were driving the results, one would expect employment to increase among GKMA firms relative to more distant firms following the reform. Appendix Table A.2 shows no such differential increase, inconsistent with systematic absorption of displaced workers. This indicates that the employment decline documented above reflects genuine destruction of formal jobs rather than a compositional shift across firms.

Effects on Firms’ Formal Wage Bill and Wage Informality:— Panel A of Figure 7 shows a differential decline in the total wage bill of treated relative to untreated firms following the reform, with an average effect of 3.8 percent. This reduction operates through two complementary channels: the mechanical contraction of the wage bill arising from the decline in formal employment documented above, and deliberate wage adjustments by firms that retain their workforce but compress wages in response to higher operating costs. The presence of both channels implies that the labor cost consequences of the re-

¹⁰I consider firms in the 15 closest districts to the GKMA. The results are consistent when this threshold is extended to 20 districts.

form extend beyond headcount reductions, with firms adjusting on both the extensive and intensive margins of their wage bill.

Firms facing higher costs associated with increases in property tax liabilities may have an incentive to shift workers off formal payrolls, reducing their social security contributions and other payroll-linked obligations, while retaining their productive services through informal contracts, an adjustment margin that would be invisible to analyses relying solely on formal employment records. To analyze this margin, I construct a proxy for informal wages, as the difference between the wage bill reported on firms' corporate income tax returns and the total wage bill derived from employee-level returns. I normalize the outcome variable by the CIT wage bill, so that coefficients are interpretable as percentage-point changes.¹¹

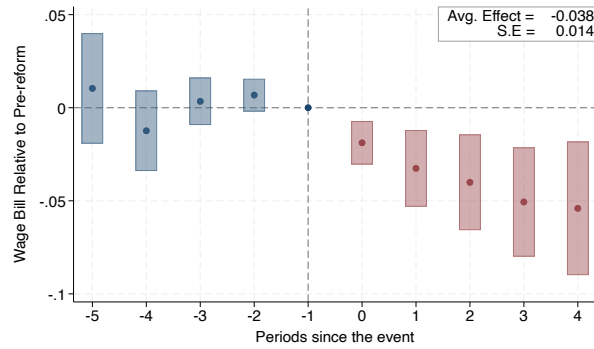
Panel B of Figure 7 shows that the share of informal wages increased by 5.1 percentage points against a pre-reform mean of 7.7 percent, and Appendix Table A.3 shows that wage informalization accounts for 14 percent of the decline in the formal wage bill.¹² This suggests that a non-trivial share of the observed decline in formal employment reflects a reallocation of labor toward informal arrangements rather than outright job destruction, implying that the aggregate employment estimates above overstate true job losses to the extent that some workers transition into informal roles rather than unemployment. Beyond this measurement implication, the informalization margin points to a broader distributional consequence of property taxation in contexts where informal labor markets are prevalent: the tax burden is partly shifted onto workers in the form of reduced labor protections and benefits associated with formal employment.

A potential concern with this approach of analyzing wage informality is that the reform may incentivize firms to inflate their reported wage bill on corporate income tax returns in order to reduce taxable income, which would generate an artificial wedge between CIT-reported and employee-level wages. If firms were systematically over-reporting labor costs in response to the reform, one would expect a broad-based inflation of other deductible expense categories as well. Appendix Figure A.13 provides evidence against this interpretation: expenses on machine repair decline while expenses on training, travel, telephone, and legal remain unchanged following the reform. Rather than reflecting strategic over-reporting, this pattern is consistent with firms cutting specific operating costs in response to higher property tax liabilities, supporting the interpretation that the wedge between CIT-reported and employee-level wages reflects genuine labor informal-

¹¹The underlying assumption is that firms report truthfully on their wage bill on the corporate income tax returns.

¹²The event-study analysis corresponding to Table A.3 is presented in Figure A.12

Panel A: Formal Wage Bill/Pre-reform Values



Panel B: Informal Wage Bill/CIT wage Bill

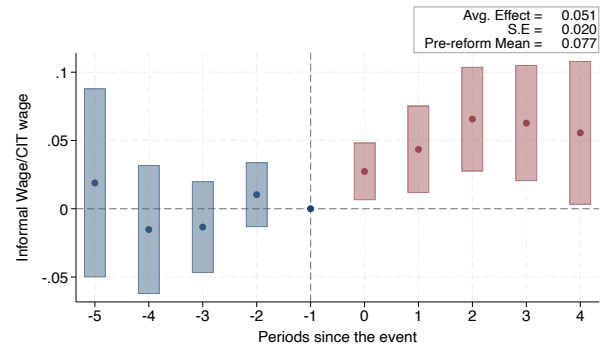


Figure 7: Property Taxes and Firms' Wage Bill

Notes: The figure presents the effects of the reform on firms' wage bill. In Panel A, the dependent variable is total wage bill, scaled by its pre-reform value. In Panel B, the dependent variable is informal wage, defined as $\frac{\text{CIT wage bill} - \text{Total wage bill from employee returns}}{\text{CIT wage bill}}$. This means coefficients can be interpreted as a percentage point changes. The horizontal axis reports the number of years relative to the reform implementation. Each dot represents the estimated treatment effect for a given year, with year -1 as the reference category. The vertical bands denote the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

ization rather than tax-motivated accounting manipulation.

V.D. Addressing the COVID-19 Threat

A potential threat to identification is that the later years of the sample overlap with the COVID-19 pandemic, which disrupted firm operations and labor markets in ways that may confound the estimated effects of the reform. The identifying assumption required for my estimates to be uncontaminated by COVID-19 is that the pandemic did not differentially affect treated and comparison firms. This assumption is plausible for three reasons. First, I restrict never-treated comparison units to firms located just outside the city boundary, which are likely to have faced similar pandemic exposure as treated firms within the city. Second, in the main analysis I use not-yet-treated firms, which includes both never-treated firms and those located within the city but treated at a later date, as the comparison group, further limiting the potential for differential COVID-19 exposure across groups. Reassuringly, the estimates remain comparable in magnitude and precision when using only never-treated firms, those located outside the city boundary, as the comparison group instead. Third, the results show that effects are concentrated among tenant businesses and that the reform led to a significant increase in rental expenses. If COVID-19 was driving the results, one would not expect COVID-19 to operate differentially through the rental cost channel, since the pandemic would not systematically raise

business rents. The fact that treatment effect heterogeneity aligns closely with exposure to higher rental costs is more naturally explained by the property tax mechanism than by pandemic disruptions.

Nevertheless, I conduct two additional exercises to assess whether the results are driven by COVID-19. First, I restrict the sample to the pre-COVID period by dropping all observations after 2018/19. Panels A, C, and E of Appendix Figure A.14 show that differential declines in employment and wages persist in the restricted sample. Although the average effects are smaller in magnitude, the event-study plots reveal that the effects were still evolving by the end of the truncated sample, suggesting that the full-sample estimates reflect the continuation of a trend that predates the pandemic rather than a COVID-induced shock, thereby confirming that the results are not an artifact of the pandemic period.

Second, I exploit the calendar-time-specific effects estimator embedded in [Callaway and Sant'Anna \(2021\)](#)'s staggered difference-in-differences framework, which recovers the average treatment effect for each calendar year separately, using only firms treated by that date. This allows me to assess whether the employment and wage effects emerge prior to the COVID-19 period or materialize only during the pandemic. Panels B and D of Appendix Figure A.14 show that employment effects were already building before the onset of the pandemic, with estimates growing in magnitude through 2018/19 and persisting into the pandemic period. Panel F shows an identical pattern for wages. The fact that the estimated effects predate the pandemic makes it unlikely that COVID-19 is the primary driver of the estimated impacts.

VI. DISTRIBUTIONAL CONSEQUENCES

The aggregate employment effects documented above mask substantial heterogeneity in how the burden of the reform is distributed across workers. In particular, if firms respond to the reform by adjusting employment at the bottom of the wage distribution—disproportionately reducing employment among low-skill, low-wage workers—the reform may disproportionately harm the most economically vulnerable workers, even when its aggregate employment effect appears moderate. As with other business taxes, the distributional consequences of property taxation thus depend critically on which workers bear the burden of firms' adjustment responses (e.g., [Risch 2024](#); [Fuest et al. 2018](#)).

To identify which types of workers are most affected by the reform, I implement two complementary firm-level analyses.¹³ First, I examine changes in the composition of em-

¹³I conduct the analysis at the firm level rather than the worker level, as workers are uniquely identified

ployment by tracking how the share of workers in each pre-reform wage quartile, relative to pre-reform total employment, evolves following the reform. Formally, the outcome variable is $\frac{E_{q,t}}{E_0}$, where $E_{q,t}$ denotes the number of workers in quartile q at time t and E_0 denotes pre-reform total employment.¹⁴ Second, I analyze how the within-firm wage distribution evolves in treated relative to untreated firms following the reform. In both analyses, I restrict the sample to firms that employed at least five workers in the pre-reform period and remained active throughout the sample period, to ensure sufficient support across the wage distribution. All estimates follow the staggered difference-in-differences approach of [Callaway and Sant'Anna \(2021\)](#).

Figure 8 presents the effects of the reform on employment composition across the pre-reform wage distribution. The reform led to a differential decline in the share of workers in the top quartile among treated relative to untreated firms, accompanied by a corresponding increase in the share of workers in the 50th–75th percentile. The absence of pre-reform differential trends in Panel A supports a causal interpretation of these patterns. On average, the share of workers in the top quartile declined by 2.8 percentage points, while the share in the 50th–75th percentile increased by 2.2 percentage points (Panel B). Effects on the bottom quartile and the 25th–50th percentile are not statistically significant, although the point estimates are positive.

Two mechanisms could account for this pattern. First, firms may substitute high-earning workers with lower-wage workers of relatively similar productivity, reducing their wage bill without proportionally contracting their workforce. Second, wages at the top of the distribution may have fallen more sharply than those at the bottom, mechanically shifting workers from the top quartile into the third quartile without any change in employment composition itself.

In the second analysis, I investigate the wage channel by examining how the within-firm wage distribution evolves before and after the reform. Specifically, I track the 20th, 40th, 60th, and 80th percentiles of the within-firm wage distribution in treated relative to untreated firms, scaling each percentile by the pre-reform total formal wage bill to allow interpretation of the estimates as percentage-point changes relative to the pre-reform wage bill. Appendix Figure A.15 shows that while coefficient estimates are negative across the entire distribution, the effects are stronger at the top: the 60th percentile of wages declines by 0.1 percentage points relative to the pre-reform wage bill, while the 80th percentile of wages declined by 0.2 percentage points, and these are the only esti-

only at the firm level in the data.

¹⁴The focus on the quartiles reflects sample size limitations.

mates that are statistically significant.¹⁵ This evidence points to within-firm wage compression as an important adjustment mechanism.

Taken together, these results indicate that the burden of higher property tax liabilities falls disproportionately on higher-earning workers within firms. The reform thus compressed the within-firm wage distribution, reducing inequality between low- and high-wage workers, a pattern that is progressive in this narrow sense. These findings relate closely to [Risch \(2024\)](#), who shows that a disproportionate share of the tax burden on business owners is ultimately borne by relatively high-earning employees. While property taxes differ from business owner taxes in their statutory incidence, both share the feature of being levied outside of direct business income, suggesting a common transmission channel through which firm-level cost shocks are absorbed along the upper tail of the within-firm wage distribution.

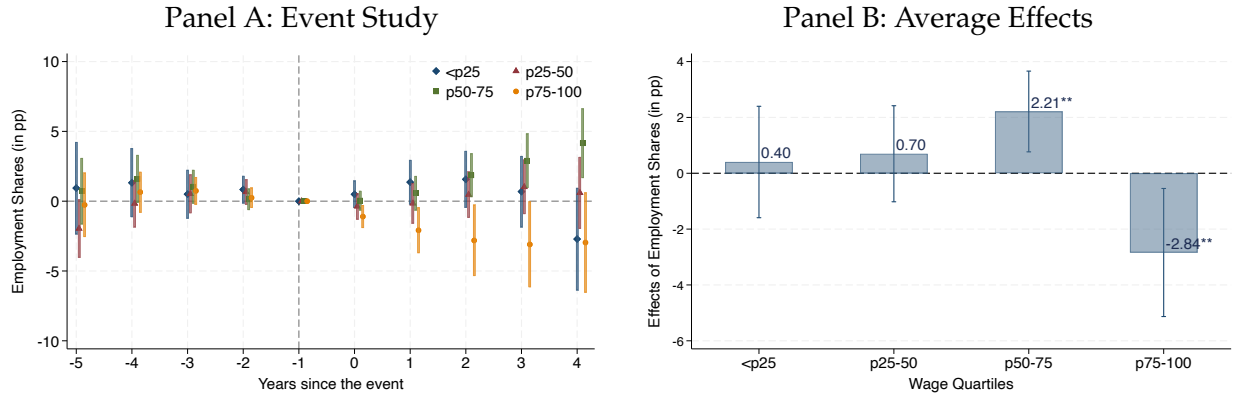


Figure 8: Effects of the Reform on Employment Composition

Notes: The figure presents the effects of the reform on within-firm employment composition. The dependent variables are employment shares in each quartile, relative to pre-reform total employment, i.e., $\frac{E_{i,b,t}}{E_{i,0}} \times 100$, where $E_{i,q,t}$ denotes firm i 's number of workers in quartile q at time t and $E_{i,0}$ denotes firm i 's total formal employment in the pre-reform period. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to pre-reform total employment. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B presents the average effects. The regression specifications include fixed effects for sector, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

¹⁵Appendix Figure [A.16](#) shows a consistent pattern for the 25th, 50th, and 75th percentiles.

VII. HETEROGENEITY ANALYSIS ON EMPLOYMENT AND FIRM SURVIVAL

VII.A. *Heterogeneity by Profitability and Leverage*

To better understand the underlying mechanisms, I examine heterogeneity by baseline profitability and leverage. Property taxes increase fixed operating costs and thereby reduce profitability; firms with thin operating margins may be pushed into losses, increasing the likelihood of firm exit or employment adjustment. To assess this channel, I first examine effects among loss-making firms. I then partition the sample into terciles and examine how the effects vary across the distribution of pre-reform profitability.¹⁶ Figure 9 shows that the magnitude of both employment and survival effects declines monotonically with profitability. The event-study plots in Appendix Figures A.19 and A.23 show that this pattern is not driven by pre-trends. The finding that survival effects are strongest among loss-making firms further rules out the possibility that the overall survival effects are driven by evasion responses to corporate income tax, since such firms have zero corporate income tax liability.

Turning to leverage, Figure 9 reveals a U-shaped pattern in employment and survival effects across the leverage distribution. The effects are statistically significant only at the two tails, among firms with very low and very high leverage, while moderately leveraged firms are largely insulated. Appendix Figures A.20 and A.24 confirm that these patterns are not driven by differential pre-trends, supporting the causal interpretation of these patterns. This U-shape points to two distinct vulnerability channels operating at opposite ends of the distribution. High-leverage firms are exposed through a financial distress channel: higher property tax obligations tighten liquidity constraints associated with existing debt service commitments, compressing already thin repayment margins and amplifying the probability of distress. At the other end, very low-leverage firms are exposed through a financial exclusion channel: absent meaningful access to credit markets, these firms cannot borrow to smooth the shock and are instead forced to adjust on real margins, cutting employment or exiting altogether. Firms with moderate leverage, by contrast, retain sufficient unused debt capacity to absorb the shock through additional borrowing, explaining their relative insulation. This U-shaped pattern of incidence suggests that it is financial flexibility, rather than financial strength per se, that shields firms from cost shocks.

Appendix Figure A.25 provides direct evidence consistent with this mechanism. Moderately leveraged firms in treated areas experience a differential increase in debt relative to their counterparts in untreated locations following the reform, consistent with these

¹⁶Profit margins are defined as the ratio of profits (EBIT) to revenue.

firms drawing on available borrowing capacity to smooth the shock. No such borrowing response is observed among firms at either tail of the leverage distribution, precisely those for whom the employment and survival effects are most pronounced, reinforcing the interpretation that access to borrowing capacity separates insulated firms from vulnerable ones.

Taken together, the heterogeneity patterns by profitability and leverage point to financial constraints and financial distress as key mechanisms underlying the reform's effects.

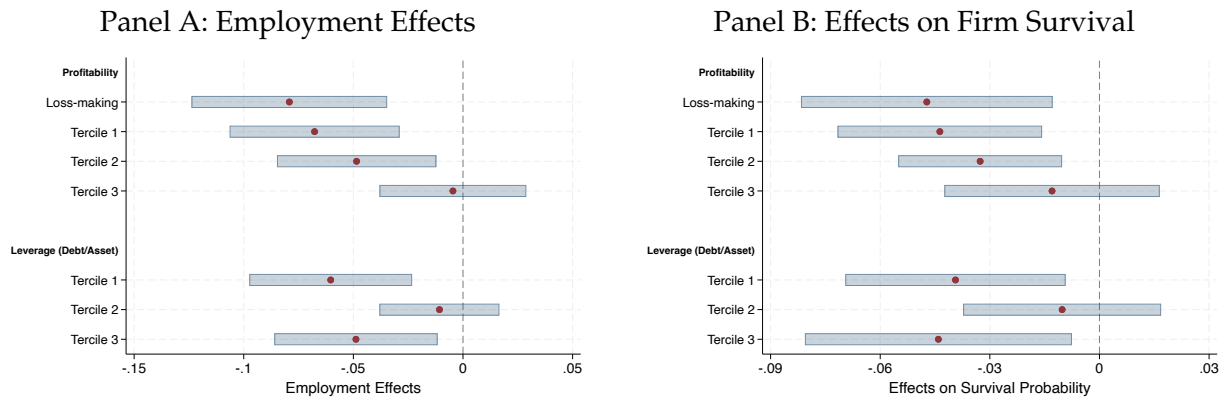


Figure 9: Employment and Survival Effects: Heterogeneity by Profitability and Leverage

Notes: The figure shows the heterogeneous effects of the reform on firm survival and employment. Each dot represents a coefficient from a separate regression. Firms are distinguished by pre-reform profitability (EBIT/revenues) and leverage (debt-to-asset ratio). The regression specifications include fixed effects for industry, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

VII.B. Heterogeneity by Occupancy Status: Owner-Occupiers vs. Tenants

The effects of property taxation on firms are likely to vary by occupancy status. Owner-occupiers bear the tax directly as part of the cost of occupying commercial space, whereas tenant firms are affected only insofar as landlords transmit the burden through higher rents. To examine this dimension, I classify firms as tenants if they report positive rental expenses in the pre-reform year and as owner-occupiers if they report positive land and building values on their CIT returns. Because some firms report both rental expenses and positive building values, these categories are not mutually exclusive. For firms reporting neither rental expenses nor building values, I assign occupancy status based on pre-reform capital intensity, the ratio of fixed assets to total assets, classifying firms below the median as tenants and those above as owner-occupiers. As a robustness check, I show that the results are qualitatively similar when firms with imputed occupancy status are excluded.

Panel A of Figure 10 shows that while the reform reduced formal employment for both tenant and owner-occupier firms, the effect is somewhat larger among tenant firms. Similarly, Panel B indicates that the negative effects on firm survival are concentrated among tenant firms. Appendix Figures A.21 and A.17 confirm that these patterns are not driven by pre-trends. This pattern contrasts with the standard neoclassical prediction, which would predict little to no effect among tenant firms. The weaker effects on either of these outcomes among firms that report owning land and buildings may reflect firm size and financial capacity: such firms are likely larger and better able to absorb higher property tax liabilities by adjusting along other margins. Consistent with this interpretation, heterogeneity by firm size points in the same direction. Employment and survival effects are only significant among the smallest firms, those with fewer than five workers (Appendix Figure A.2).

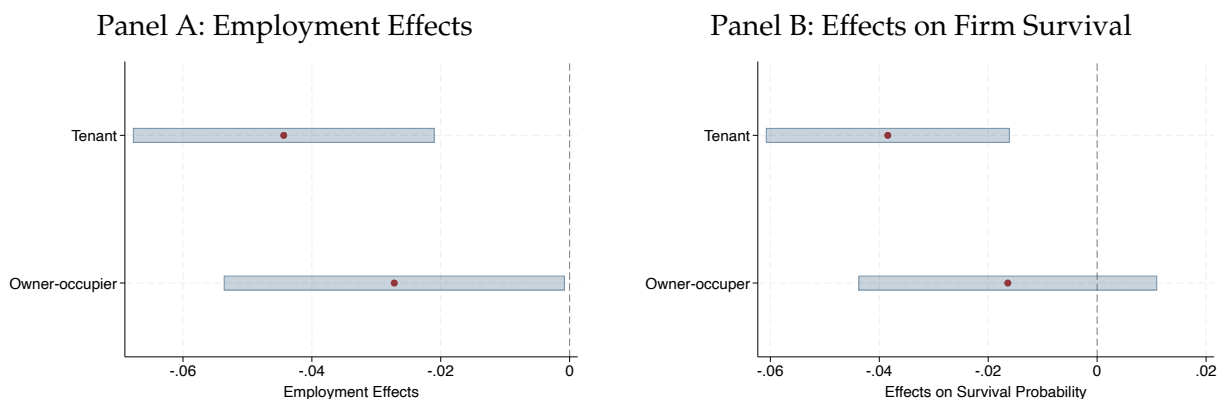


Figure 10: Employment and Survival Effects by Occupancy Status

Notes: The figure shows the heterogeneous effects of the reform on firm survival and employment. Each dot represents a coefficient from a separate regression. Firms are distinguished based on their pre-reform occupancy status (tenant business or owner-occupier). The regression specifications include fixed effects for industry, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

VIII. BUSINESS RENTAL EXPENSES

The heterogeneity results in Section VII.B suggest that the rental cost channel plays an important role in transmitting property tax burdens to firms. To quantify the importance of this mechanism, I estimate the effect of the reform on business rental expenses and calculate the implied pass-through rate. I use an unbalanced panel of firms that reported positive rental expenses in the pre-reform year and estimate the effect of the reform on rental expenses normalized by their pre-reform values.

The results are presented in Figure 11. Prior to the reform, rent expenses for firms in locations exposed to increases in property tax liabilities and those in unaffected areas followed similar trends, supporting the parallel trends assumption. Following the reform, firms in treated locations experienced a differential increase in rent expenses relative to those in untreated areas. The average effects presented in Table 4 show that the reform led to a 6.5 percent increase in business rental expenses (column 4). Appendix Figure A.26 shows that the results are consistent when restricting the sample to a balanced panel of firms that reported positive rental expenses in the pre-reform period, although the coefficient estimate is larger in magnitude.

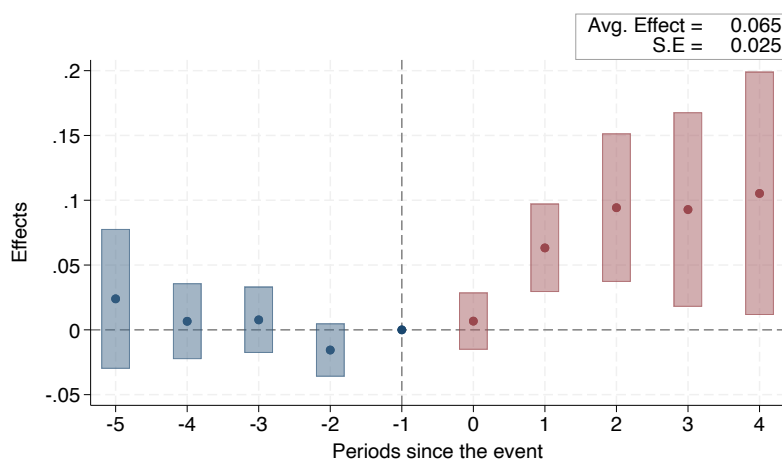


Figure 11: Effects of the Reform on Business Rental Expenses

Notes: The figure shows the dynamic effects of the reform on business rental expenses. The dependent variable is rent costs scaled by its pre-reform value, implying that the coefficients can be interpreted as percentage changes. The effects are estimated on an unbalanced panel of firms that reported paying rent in the pre-reform period. The horizontal axis shows the number of years since the policy. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered difference-in-differences approach, following Callaway and Sant'Anna (2021), with not-yet treated units used as the comparison group. Standard errors are clustered at the firm level. Appendix Figure A.26 shows the effects using a balanced panel while Figure A.27 shows the effects using never treated units as the comparison group.

Implied Pass-Through Rate:— To interpret the economic magnitude of the reform's effects on rental expenses, I compute an implied pass-through rate using the reduced-form estimate. As noted in Section ??, I do not observe firm-specific property tax liabilities, which makes it impossible to estimate pass-through at the individual firm level.

Given these constraints, I estimate the pass-through rate for the population of formal

firms, i.e., those registered for corporate income tax, operating in Kampala.¹⁷ Conceptually, this approach treats the reform as an aggregate shock to the cost of occupying commercial property and asks what share of the resulting increase in property tax liabilities is transmitted to formal tenant firms through higher rents. I operationalize this by scaling the total increase in rental expenses among CIT-registered tenant firms by the aggregate increase in property tax liability induced by the reform. Formally, I define the pass-through rate as: $\frac{\hat{\beta}_{rent} * \bar{R} * N_{renters}}{\Delta T^{City}}$, where $\hat{\beta}_{rent}$ is the estimated average treatment effect on rental expenses in percent, $\bar{R}$ is the average pre-reform rent expense, $N_{renters}$ is the number of CIT-registered firms that reported positive rental expenses in the pre-reform period, and ΔT^{City} is the reform-induced change in citywide commercial property tax liabilities.¹⁸

Results in Table 4 (second row from the bottom) indicate that a UGX 1 increase in aggregate property tax revenue is associated with a UGX 0.53 increase in total rental expenses paid by CIT-registered tenant firms. This measure captures the share of the aggregate increase in property tax liabilities borne by formal tenant firms through higher rents. It should be interpreted as a pass-through rate to the formal sector specifically, rather than a citywide rate encompassing all commercial tenants, since the CIT-filing population does not cover the universe of businesses operating in Kampala. Even within this population, the estimate is likely a lower bound on the true pass-through to formal tenant firms, since the denominator reflects the total property tax increase levied on all commercial properties citywide, including those occupied by non-filing businesses. To the extent that the denominator overstates the tax increase attributable to properties occupied by formal tenant firms, the resulting pass-through rate is biased downward. Moreover, the estimate is an aggregate measure that pools across firms facing different levels of tax exposure and occupying properties of varying types and values, reflecting the average pass-through across a heterogeneous population rather than the pass-through on any individual property. It should therefore not be interpreted as a structural, firm-level pass-through rate.

Although not directly comparable to property-level pass-through estimates in the housing literature, the pass-through rate to formal firms estimated here is broadly consistent in magnitude with recent studies of property tax incidence in rental markets. These studies typically estimate pass-through at the property or unit level and, in some cases, embed the analysis in general equilibrium frameworks that allow for endogenous adjustment in rents, occupancy, and supply, finding pass-through rates ranging from \$0.50 to \$1.39

¹⁷Here, formal refers specifically to firms registered for corporate income tax with the Uganda Revenue Authority, which constitutes a subset of all businesses operating in Kampala.

¹⁸Since $\hat{\beta}_{rent}$ is estimated using an unbalanced panel that includes all pre-reform tenant firms regardless, the numerator captures the change arising from the population of CIT-registered tenant firms.

per \$1 of tax (e.g., [Baker 2024](#); [Watson and Ziv 2024](#); [Löffler and Siegloch 2021](#); [Rolheiser 2019](#)). By contrast, my estimate is a reduced-form aggregate incidence measure for CIT-registered tenant firms, pooling across heterogeneous properties, landlords, and tenant firms, and abstracting from equilibrium reallocation across locations. Despite these differences in the level of aggregation, empirical design, and institutional context, the similarity in magnitude suggests that a substantial share of the commercial property tax burden is transmitted to firms through higher rents.

My results are consistent with the “old view” of property tax incidence, which posits that the portion of the tax falling on land is borne entirely by owners, since the supply of land is perfectly inelastic, while the portion falling on improvements—i.e., structures—can be shifted to final consumers and occupants, since the supply of capital to the construction industry is relatively elastic ([Oates and Fischel 2016](#)). In my setting, it is plausible that the tax falls primarily on improvements rather than on land itself, since idle land is exempt.

Table 4: Effects of the Reform on Business Rental Expenses

	(1)	(2)	(3)	(4)
Average Effect	0.052** (0.023)	0.067*** (0.025)	0.066*** (0.025)	0.065*** (0.025)
Sector of Operation	N	Y	Y	Y
Pre-reform Size	N	N	Y	Y
Accounting Month	N	N	N	Y
Pre-reform Mean	28.68	28.68	28.68	28.68
Implied Pass-Through Rate	0.427	0.545	0.538	0.532
Observations	56761	56761	56761	56761

Note: The table presents average effects of the reform on business rental expenses. The effects are estimated on an unbalanced panel of firms that reported paying rent in the pre-reform period. The dependent variable is rent costs scaled by its pre-reform value, implying the coefficients can be interpreted as a percent change. Each column reports estimates from a different specification with an alternative set of controls. The second row from the bottom presents the pass-through rate to formal firms calculated from the expression: $\frac{\hat{\beta}_{rent} * \bar{R} * N_{renters}}{\Delta T^{City}}$, where $\hat{\beta}_{rent}$ is the estimated average treatment effect on rental expenses in percent, $\bar{R}$ is the average pre-reform rent expense, $N_{renters}$ is the number of CIT-registered firms that reported positive rental expenses in the pre-reform period, and ΔT^{City} is the reform-induced change in citywide commercial property tax liabilities. The coefficients are estimated using a staggered Difference-in-Differences approach following [Callaway and Sant’Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level. Pre-reform mean is measured in millions of Uganda Shillings.

The finding that higher property tax liabilities increase business rental expenses contrasts with neoclassical predictions, which suggest little to no pass-through of property taxes ([Mieszkowski 1972](#); [Zodrow 2001](#)). In Appendix II, I propose a framework that

rationalizes this result. The model incorporates a key institutional feature of selective enforcement: the probability of enforcement, and thus compliance, is higher when a property is occupied and generating observable economic activity, and lower when it is vacant. This feature is likely prevalent in developing country contexts, where limited administrative capacity leads tax authorities to target visibly active properties. As a result, the tax shifts from being a cost of owning property to a cost of supplying space to the rental market. In this setting, property taxes act as a marginal cost of renting out space, and standard incidence logic predicts positive pass-through to tenants.

Occupancy Status: Own or Lease?— Since the incidence of property taxes depends on occupancy status, higher tax liabilities may prompt firms to reconsider whether to own or lease their operating space. I examine this margin by classifying firms into pure owners, pure renters, and mixed firms based on pre-reform records, and tracking transitions across these categories following the reform. Appendix Figure A.28 shows that the reform did not induce complete switches between full ownership and full tenancy, unsurprising given the lumpiness and illiquidity of commercial property transactions. The more informative margin operates at the intensive margin: the reform differentially induced pure owners to transition into mixed arrangements, supplementing owned premises with leased space, while pure renters were less likely to acquire property and mixed firms were more likely to unwind their ownership stake. These findings indicate that property tax reform reshapes occupancy decisions primarily through gradual adjustments along the intensive margin rather than complete tenure switching, with owner-occupiers incorporating leased space into their operations rather than divesting owned property outright.

IX. IMPLICATIONS FOR CENTRAL GOVERNMENT TAX REVENUE

The firm-level effects documented in Sections V.A, V.B, and V.C have direct implications for central government tax revenues. To the extent that the reform reduces firm profitability, induces firm exit, and contracts formal employment and wages, it erodes the tax bases of both the corporate income tax and the employee income tax (Pay-As-You-Earn, PAYE), potentially offsetting part of the additional property tax revenue raised by the local government. This fiscal externality represents a form of vertical tax competition: when a sub-national government raises property taxes, it may inadvertently compress the tax bases of taxes administered at the national level, generating a negative fiscal spillover that is not internalized by the local authority setting the tax rate but is consequential for

consolidated public revenues.¹⁹ Whether and to what extent such spillovers materialize is ultimately an empirical question. To examine this question directly, I study how central government tax revenues evolve around the reform.

Figure 12 presents the event-study estimates and average effects on central government tax revenues, measured in millions of Uganda shillings. I examine three outcomes in turn: corporate income tax payments (Panel A), employee income tax (PAYE) payments aggregated at the firm level (Panel B), and the combined revenue from both taxes (Panel C).²⁰ Despite inducing firm exit and reducing profits among a subset of firms, the reform had no statistically significant effect on corporate income tax revenues. This result is consistent with the earlier finding that the firms most likely to exit were precisely those with little or no corporate income tax liability—loss-making and least profitable firms that contributed little to the corporate tax base even before the reform.

The picture is different for employee income tax revenues. Panel B reveals a differential decline in PAYE receipts from treated relative to untreated firms following the reform, driven by declines in formal employment and wages, as well as increases in wage informality documented in Section V.C. These labor market responses translate directly into a smaller formal wage bill from which PAYE is withheld, reducing central government receipts.

Combining both taxes, Panel C shows that total central government revenue collected from treated firms declined by an average of UGX 0.5 million annually relative to untreated firms, against a pre-reform mean of UGX 7.8 million—a decline of 6.4 percent. As reported in Appendix Table A.7, this is equivalent to UGX 0.26 of lost central government revenue for every UGX 1 increase in property tax revenues, suggesting that the fiscal gains at the local level are substantially offset by revenue losses at the central level.

X. CONCLUSION

This paper studies the economic effects of commercial property taxation, providing new evidence on its effects on firm-level real activity, tax incidence, and vertical fiscal externalities. I exploit a natural experiment in Kampala, the capital city of Uganda, where property tax liabilities increased following a revaluation and mass property enumeration exercise, while surrounding areas were unaffected. This setting allows me to estimate causal effects using a difference-in-differences identification strategy, using novel admin-

¹⁹This vertical fiscal externality is distinct from the horizontal tax competition literature, which focuses on competition between jurisdictions at the same level of government (see, e.g., [Wilson 1999](#)).

²⁰I aggregate employee taxes at firm level because the data does not allow me to follow workers over time, as workers are only uniquely identified at firm level.

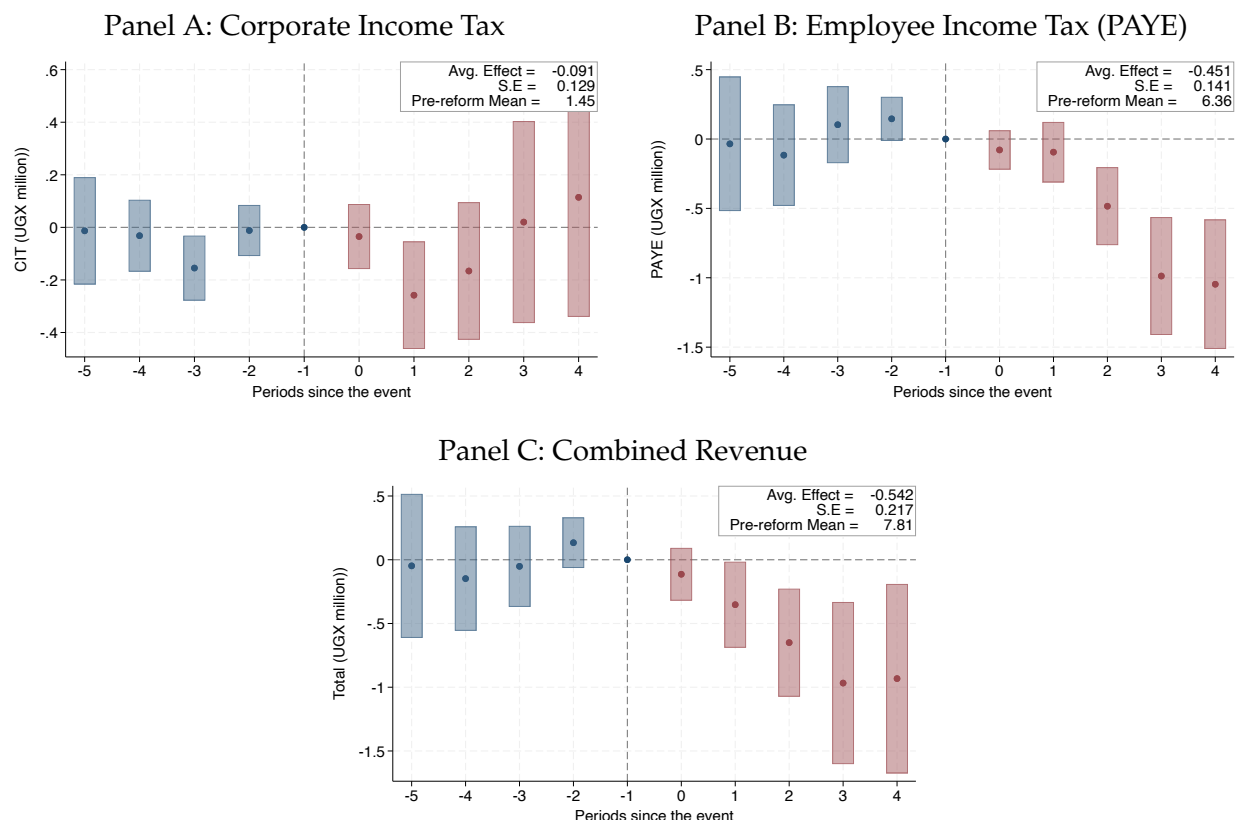


Figure 12: Implications for Central Government Tax Revenue

Notes: The figure presents the effects of the reform on central government tax revenue. All analysis is conducted at the firm level. The outcome variable is in millions of Uganda shillings (UGX), constant 2016/17. Panel A shows the effects on corporate income tax, Panel B shows the effects on employee tax revenue (PAYE), while Panel C shows the effects on the combined revenue. Each dot denotes the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for pre-reform industry-size bins and accounting month. Standard errors are clustered at the firm level.

istrative data from corporate income tax and employee tax returns. I document four main findings.

First, I show that while the reform had no significant effect on revenues, it reduced firms' profitability, investment, employment, and survival probabilities, while increasing wage informality. These effects are highly heterogeneous across firm size. Small firms experience declines in both revenue and profits, whereas large firms see revenue gains with no change in profits, consistent with greater market power and the ability to pass costs through to consumers. This heterogeneity in pass-through capacity implies that the burden of property taxation falls disproportionately on smaller firms, while larger firms remain relatively insulated. Further heterogeneity analysis reveals that pre-reform prof-

itability and leverage are important moderators: effects are strongest among less profitable firms and decline monotonically with profitability. Moreover, access to borrowing plays a critical role in enabling firms to absorb cost shocks, insulating moderately leveraged firms from the most adverse real effects.

Second, the reform had progressive distributional consequences within firms. Workers at the top of the earnings distribution were disproportionately negatively affected. In treated firms, the share of workers in the top quartile declined relative to untreated firms, with a corresponding increase in the share in the 50th–75th percentile range. There is no evidence of a significant change at the bottom of the distribution. Consistent with these compositional shifts, wages declined significantly at the top, while remaining unchanged at the bottom. Taken together, these findings indicate a reduction in within-firm wage inequality, consistent with a degree of skill downgrading.

Third, I provide evidence on the pass-through of property taxes into business rental costs. The reform increased business rental expenses by 6.5 percent, implying a pass-through rate of 0.53 among formal tenant firms, contrary to the neoclassical prediction that landlords bear the full burden. To rationalize this finding, I propose a framework that incorporates a key institutional feature of selective enforcement: the probability of enforcement, and thus compliance, is higher when a property is occupied and generating observable economic activity, and lower when it is vacant. This mechanism links occupancy to effective tax liability, thereby allowing landlords to shift a substantial share of the tax burden onto tenants.

Finally, I document evidence of a vertical fiscal externality. Reduced-form estimates show that the reform led to a decline in central government income tax revenue. Back-of-the-envelope calculations indicate that for every UGX 1 increase in property tax revenue, approximately UGX 0.26 is lost in central government revenue, suggesting that local fiscal gains are substantially offset by losses at the central level. I further show that increased wage informality is a primary channel through which this externality operates.

Taken together, these findings reveal that commercial property taxation in low-income urban settings generates consequences that extend well beyond the property market itself, reshaping firm behavior, altering the distribution of wages within firms, and eroding the tax base of other levels of government. The results underscore a fundamental tension in decentralized fiscal systems: local governments pursuing legitimate revenue objectives can impose costs that are borne not only by firms and workers within their jurisdiction, but also by the central government through diminished income tax collections. For policymakers, the central implication is that the design and evaluation of sub-national tax instruments cannot proceed in isolation. Effective fiscal policy requires coordination across

levels of government and careful attention to the behavioral margins through which tax burdens are transmitted, absorbed, and ultimately shifted onto the most vulnerable actors in the economy.

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I. APPENDIX: ADDITIONAL FIGURES AND TABLES

I.A. Additional Figures

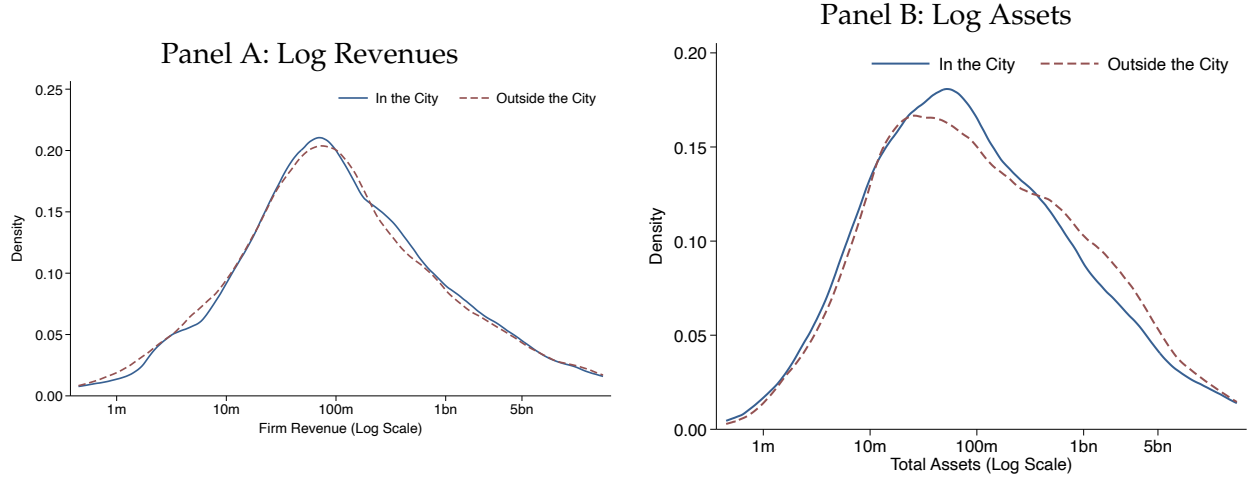


Figure A.1: Firm Size Distributions by Locations

Notes: This figure plots the size distributions of active pre-reform firms on the log scale for firms located inside the city (treated) and outside the city (comparison). Panel A shows the distribution of firm revenues, while Panel B shows the distribution of assets.

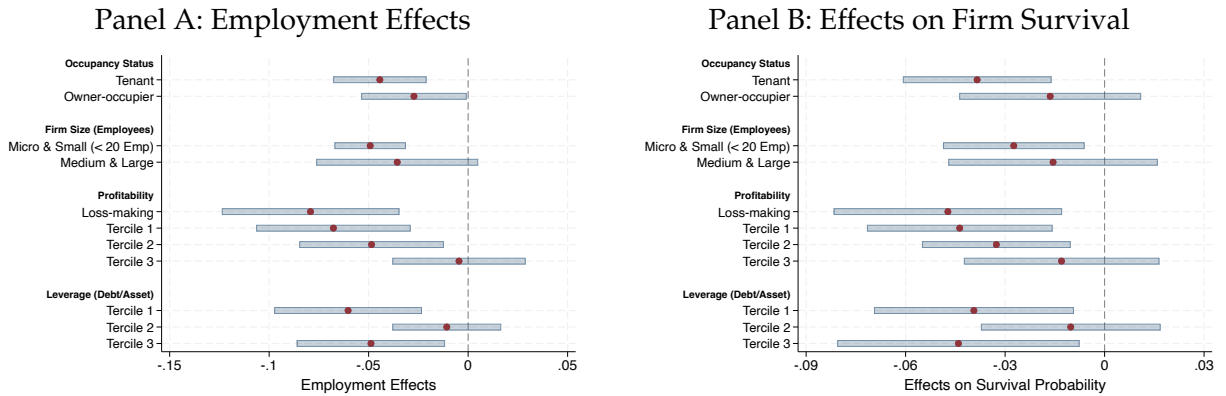


Figure A.2: Employment and Survival Effects by Firm Characteristics

Notes: The figure presents heterogeneous effects of the reform on firm survival and employment. Each dot represents a coefficient estimate from a separate regression. Firms are categorized according to pre-reform characteristics based on (i) occupancy status (tenant business or owner-occupier); (ii) firm size (measured by the number of formal employees); (iii) profitability (EBIT/revenues); and (iv) leverage (debt-to-asset ratio). The regression specifications include fixed effects for industry, pre-reform firm size, and accounting month. Standard errors are clustered at the firm level.

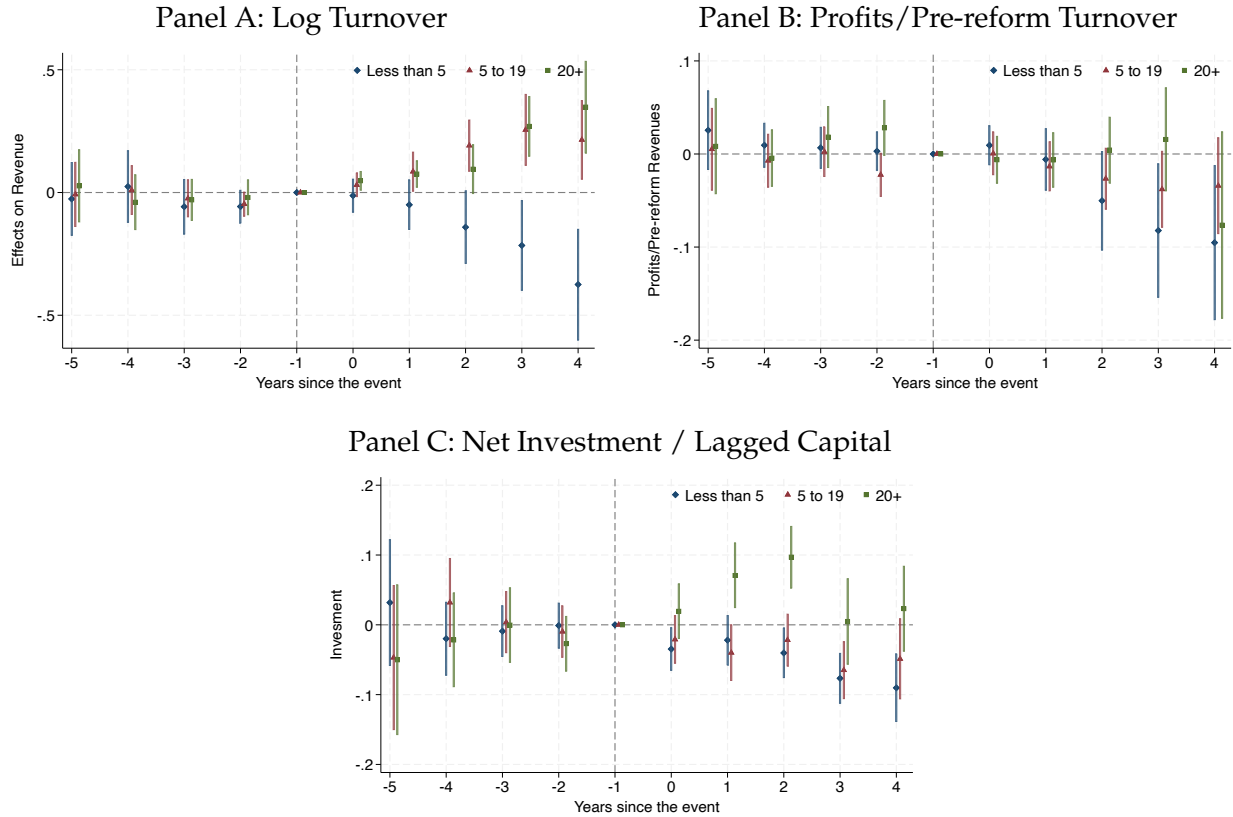


Figure A.3: Effects on Revenues, Profits, and Investment: *Heterogeneity by Firm Size*

Notes: The figure presents the effects of the reform on revenues, profits, and investment across firms of different sizes (measured by the number of formal workers). Revenues are expressed in logarithms. Profits are defined as earnings before interest and taxes (EBIT), computed as sales less the cost of sales and operating and administrative expenses. Profits are then scaled by pre-reform revenues. The investment rate is defined as the annual change in net fixed assets, scaled by lagged fixed assets. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform firm size, where applicable. The coefficients are estimated using the staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

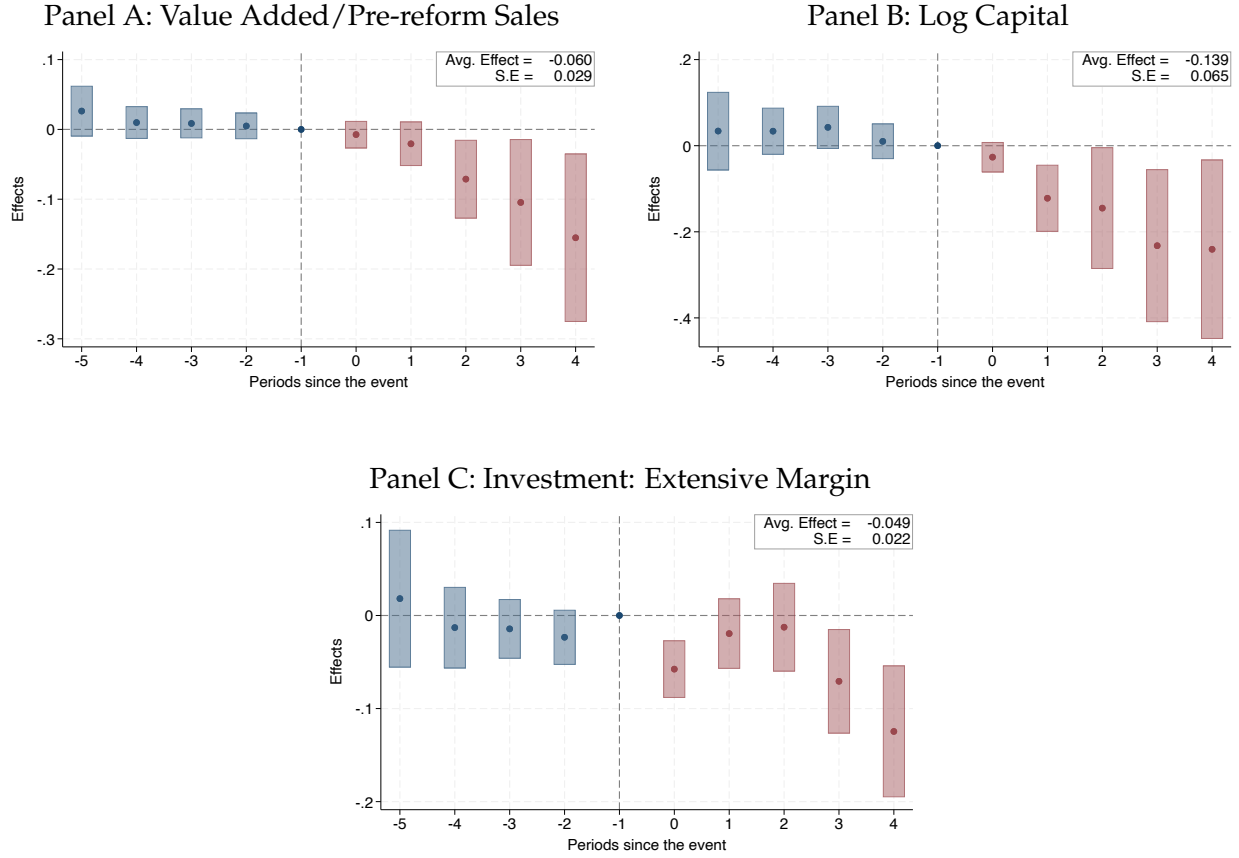


Figure A.4: Effects on Value Added, Capital, and Investment at the Extensive Margin

Notes: The figure presents the dynamic effects of the reform on firms' value added, log capital, and investment at the extensive margin. Value added is defined as revenues minus the cost of sales, scaled by pre-reform sales. Capital is measured as the book value of net fixed assets. Investment is measured as the annual change in fixed assets. At the extensive margin, the outcome is an indicator equal to one if the firm reports positive investment and zero otherwise. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform size bins. The coefficients are estimated using the staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

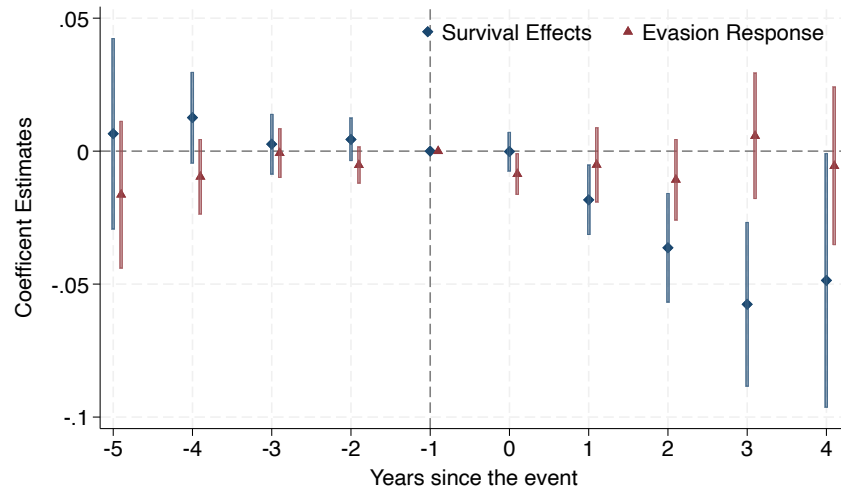


Figure A.5: Effects on Firm Survival versus Evasion Responses

Notes: The figure presents the dynamic effects of the reform on firm survival, distinguishing firm exit from evasion responses. The outcome variable is an indicator equal to one if the firm operated at time t , zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform size bins. The coefficients are estimated using the staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

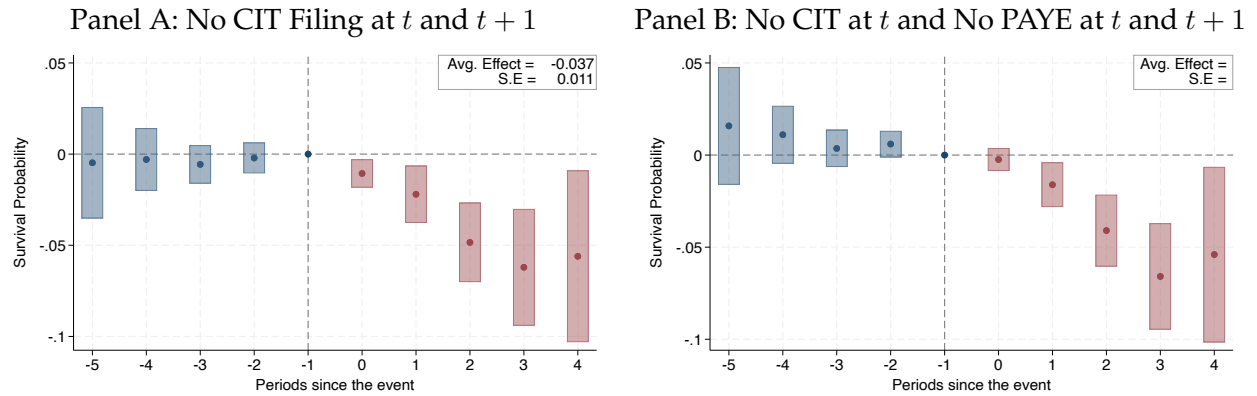


Figure A.6: Property Taxation and Firm Survival: Robustness to Alternative Exit Definitions

Notes: The figure presents robustness estimates for the effects of the reform on firm survival. In Panel A, I define exit as the absence of a corporate income tax (CIT) return for at least two consecutive years. In Panel B, I restrict exit to firms that have no CIT return in year t and no records in the employee tax returns in both years t and $t + 1$. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using the staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

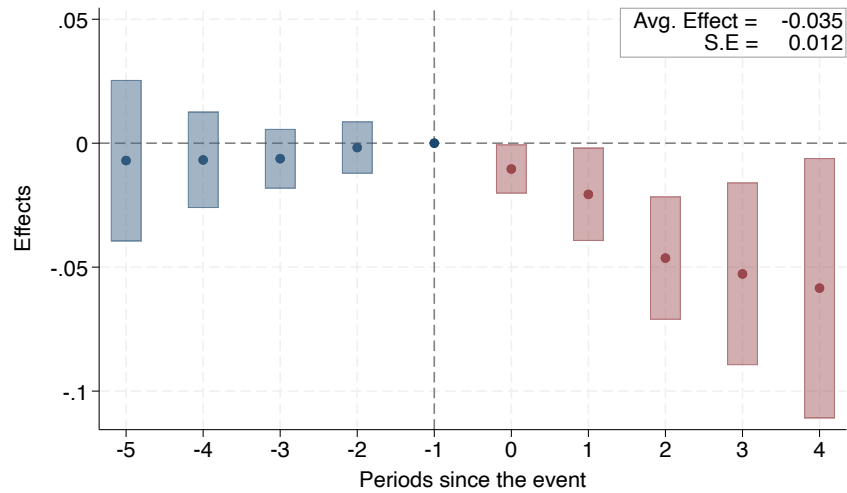


Figure A.7: Property Taxation and Firm Survival: Firms Present in the First Sample Year

Notes: The figure presents the dynamic effects of the reform on firm survival, restricting the sample to firms active in the first year of the sample rather than firms active in the year immediately preceding the reform. The outcome variable is an indicator equal to one if the firm operated at time t , zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bands indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using the staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

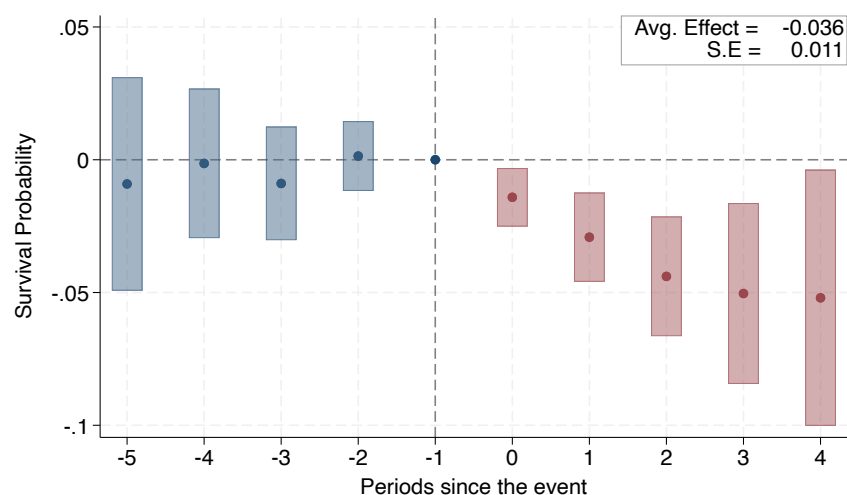


Figure A.8: Property Taxation and Firm Survival: Never-Treated Firms as the Comparison Group

Notes: The figure presents event-study estimates of the effects of the reform on firm survival, using never-treated units as the comparison group rather than not-yet-treated units. The outcome variable is an indicator equal to one if the firm operated at time t , zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform, where negative values correspond to pre-treatment periods. Each dot represents the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bands indicate the corresponding 90 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using the staggered difference-in-differences approach following [Callaway and Sant'Anna \(2021\)](#), using never-treated units as the comparison group. Standard errors are clustered at the firm level.

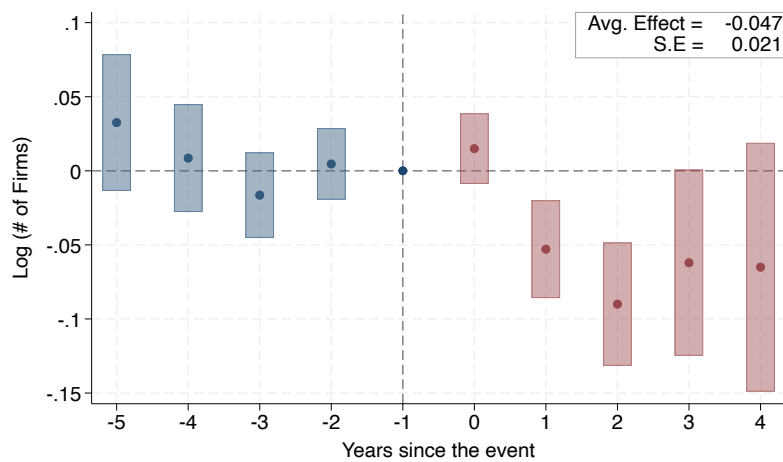


Figure A.9: Effects of the Reform on the Number of Operating Firms

Notes: The figure presents the effects of the reform on the number of operating firms in each county-industry cell. The data are aggregated to the county-industry level using four-digit industry classifications, and the outcome variable is the log number of firms in each cell. Coefficients are estimated using the staggered difference-in-differences estimator of [Callaway and Sant'Anna \(2021\)](#), with not-yet-treated units serving as the comparison group. Each point estimate represents the average treatment effect for a given event-time period relative to the reform, with period -1 as the reference category. The vertical bands denote the corresponding 95 percent confidence intervals. Standard errors are clustered at the county-industry level.

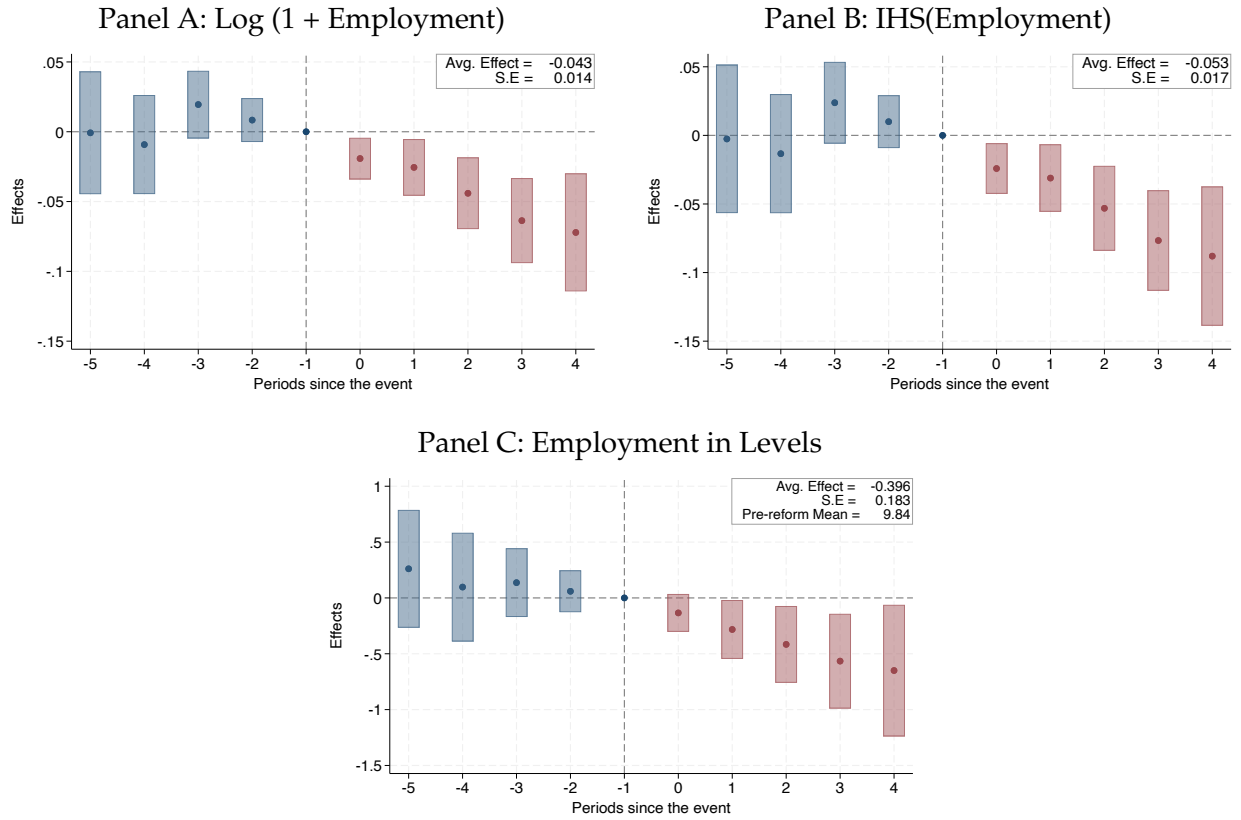


Figure A.10: Effects on Formal Employment: Logs, IHS, and Levels

Notes: The figure presents event-study estimates of the effects of the reform on firms' formal employment. In Panel A, employment is expressed in logarithms, i.e., $\log(1+\text{employment})$. In Panel B, employment is transformed using the inverse hyperbolic sine (IHS) transformation. Panel C presents the effects in levels, where the scale on the vertical axis is the number of jobs. Each dot represents the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars represent the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for pre-reform industry-size bins and accounting month. Standard errors are clustered at the firm level.

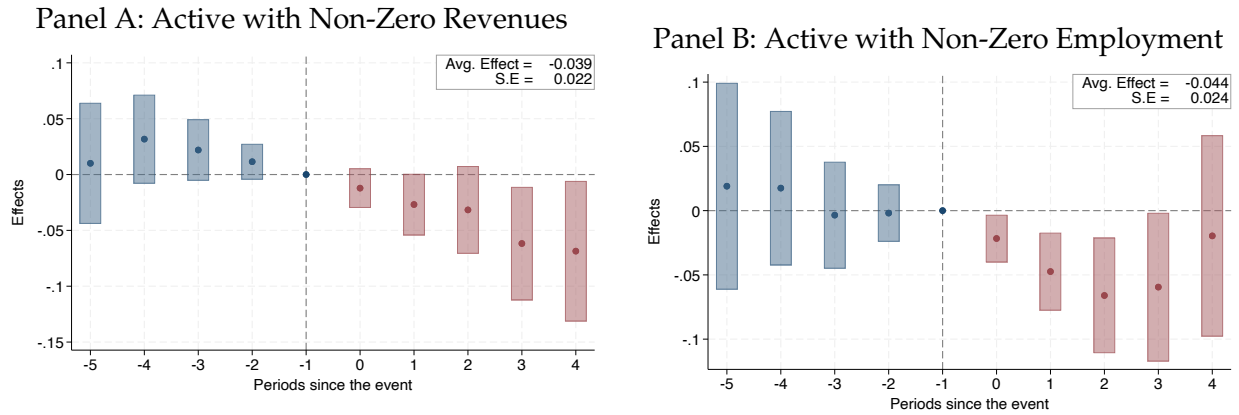


Figure A.11: Effects on Formal Employment: Firms Active Throughout the Sample Period

Notes: The figure presents the effects of the reform on formal employment, restricting the sample to firms that remain active throughout the sample period. Panel A restricts the sample to firms with non-zero revenues in all years, scaling the dependent variable by its pre-reform value. Panel B restricts the sample to firms with non-zero workers in all years, expressing the outcome variable in logarithms. Each dot represents the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars represent the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform firm size, and accounting month. The coefficients are estimated using the staggered difference-in-differences estimator of [Callaway and Sant'Anna \(2021\)](#). Standard errors are clustered at the firm level.

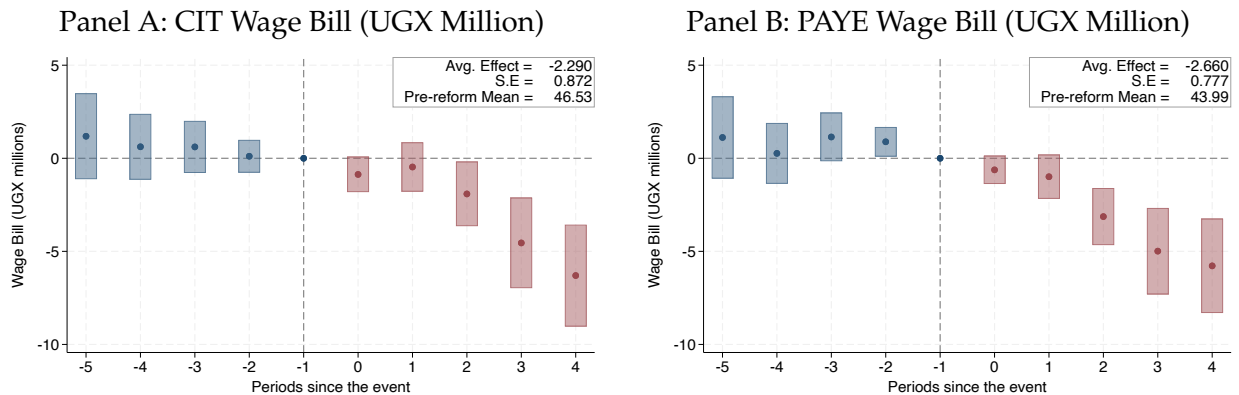
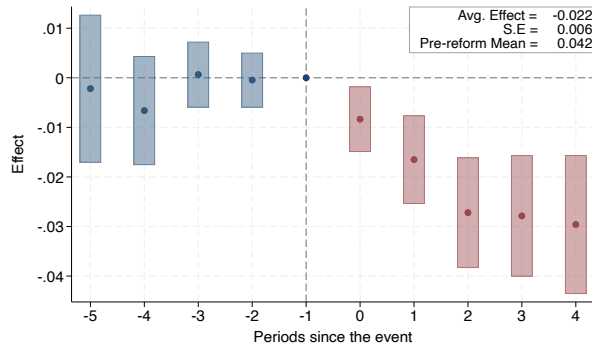


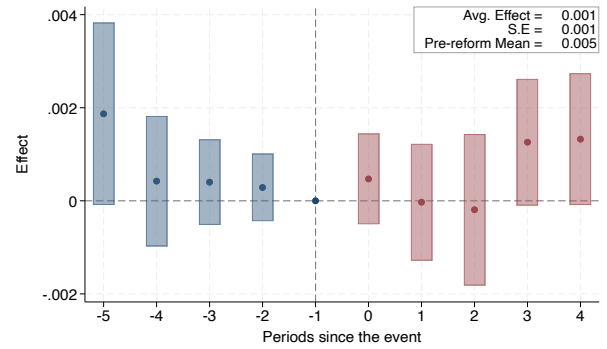
Figure A.12: Effects on the Wage Bill in Levels

Notes: The figure presents the effects of the reform on firms' wage bills. Panel A plots the effects on wage bills reported in firms' CIT returns, while Panel B reports wage bills constructed from employee tax returns. In both panels, values are expressed in UGX millions. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using the staggered difference-in-differences estimator of [Callaway and Sant'Anna \(2021\)](#). Standard errors are clustered at the firm level.

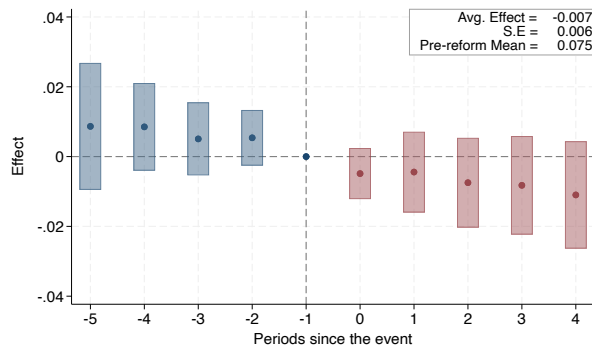
Panel A: Machine Repair/Pre-reform OPEX



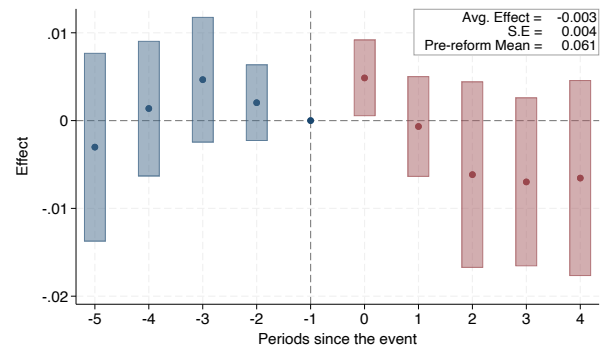
Panel B: Training Expense/Pre-reform OPEX



Panel C: Travel Expense/Pre-reform OPEX



Panel D: Phone Expenses/Pre-reform OPEX



Panel E: Legal Expenses/Pre-reform OPEX

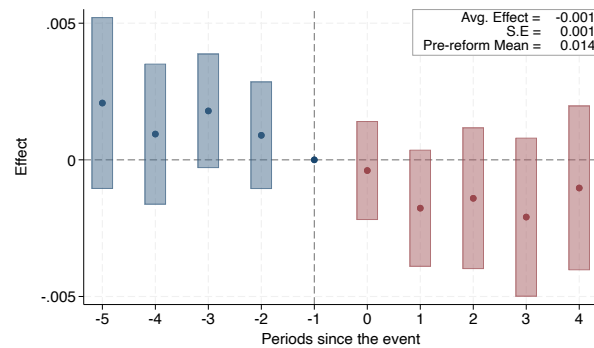
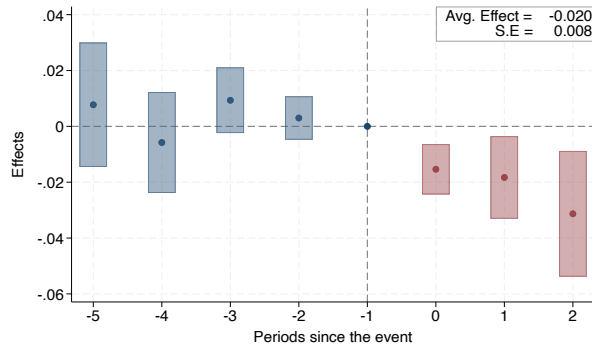


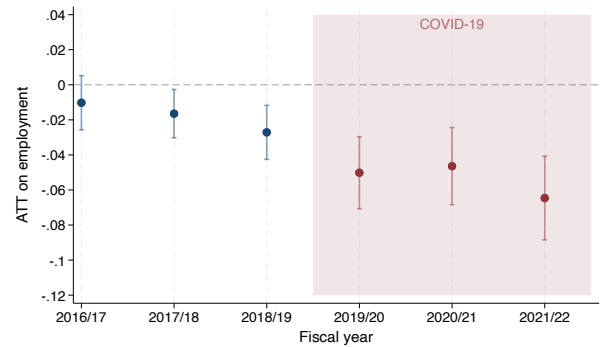
Figure A.13: Effects on Selected Categories of Operating Expenses

Notes: The figure presents the effects of the reform on selected categories of operating expenditures. In each panel, the outcome variable is scaled by pre-reform total operating expenses. Each dot represents the point estimate of the treatment effect for that year, with the reference period set at -1. The vertical bars represent the corresponding 95 percent confidence intervals. The regression specifications include fixed effects for pre-reform industry-size bins and accounting month. Standard errors are clustered at the firm level.

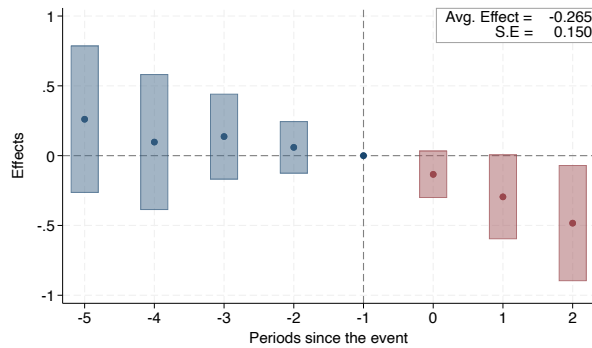
Panel A: Employment Effects: Pre-COVID Period



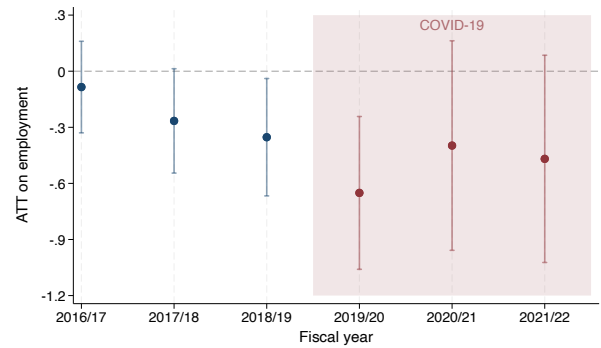
Panel B: Employment: Calendar-Time-Specific Effects



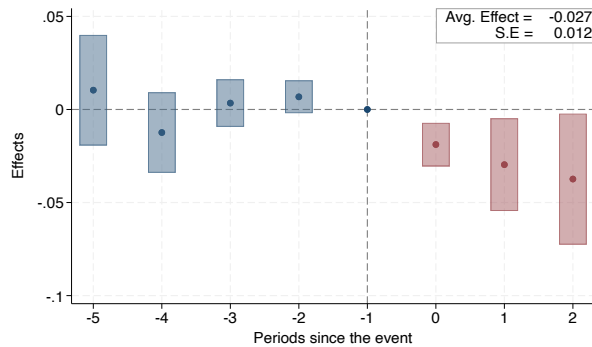
Panel C: Emp (Levels): Pre-COVID Period



Panel D: Emp (Levels): Calendar-Time-Specific Effects



Panel E: Wages: Pre-COVID Period



Panel F: Wages: Calendar-Time-Specific Effects

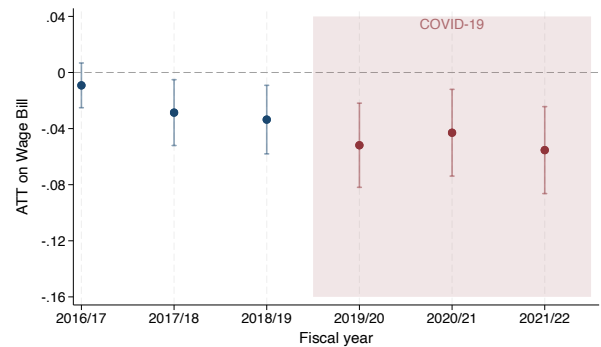


Figure A.14: Effects on Employment and Wages: Addressing COVID-19

Notes: Panels A, C, and E present event-study estimates of the effects on employment and wages, while restricting the sample to the pre-COVID period, such that the sample ends in 2018/19. Panels B, D, and F present calendar-time-specific treatment effects on employment and wages. The coefficients are estimated using a staggered difference-in-differences design following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

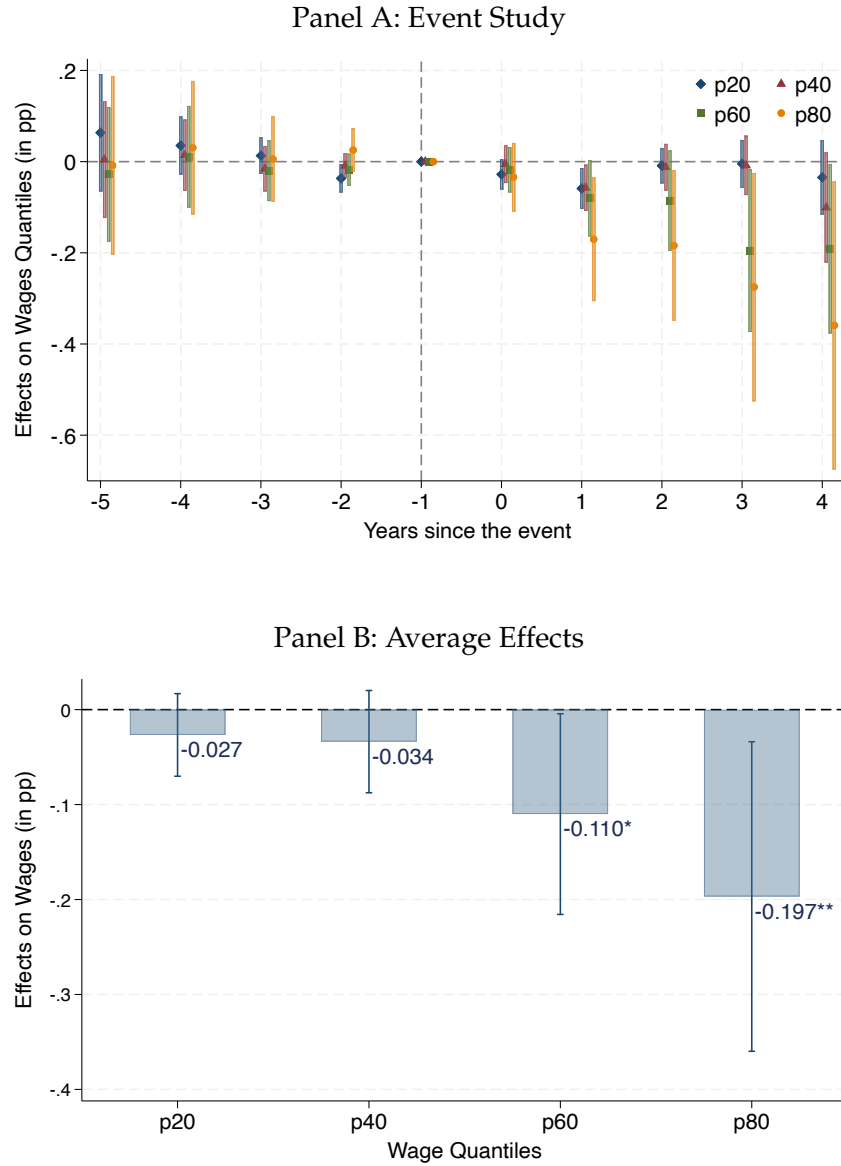


Figure A.15: Effects on Within-Firm Wage Quantiles

Notes: The figure presents the effects of the reform on within-firm wage quantiles across the wage distribution. The dependent variables are the 20th, 40th, 60th, and 80th percentiles of the within-firm wage distribution, scaled by the firm's pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i 's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i 's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the firm's pre-reform wage bill. Panel A presents the event-study estimates, with vertical bands indicating the corresponding 95 percent confidence intervals. Panel B reports the average treatment effects. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using the staggered difference-in-differences design following [Callaway and Sant'Anna \(2021\)](#). Standard errors are clustered at the firm level.

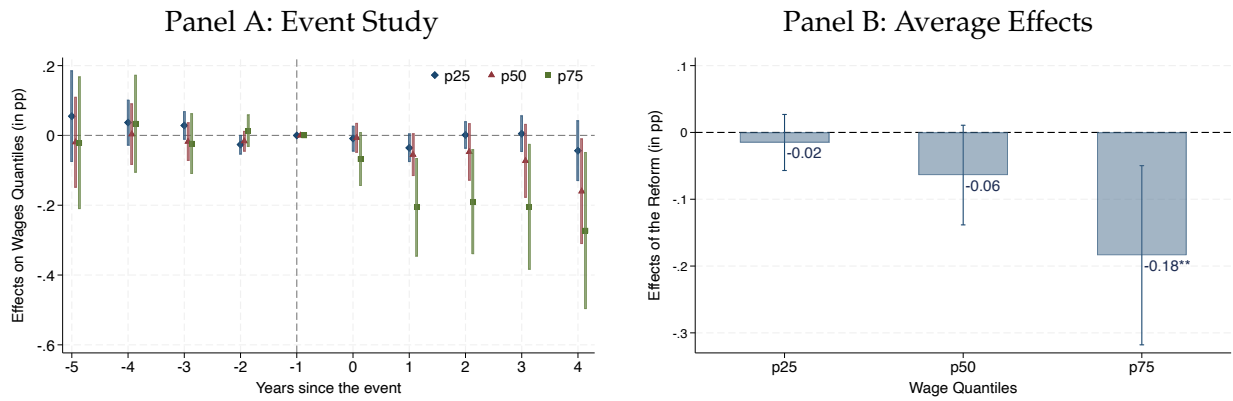


Figure A.16: Effects on Wage Quantiles Relative to the Pre-reform Wage Bill

Notes: The figure presents the effects of the reform on within-firm wage quantiles across the wage distribution. The dependent variables are the 25th, 50th, and 75th percentiles of the within-firm wage distribution, scaled by the firm's pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i 's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i 's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the firm's pre-reform wage bill. Panel A presents the event-study estimates, with vertical bars indicating the corresponding 95 percent confidence intervals. Panel B reports the average treatment effects. The regression specifications include fixed effects for sector, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

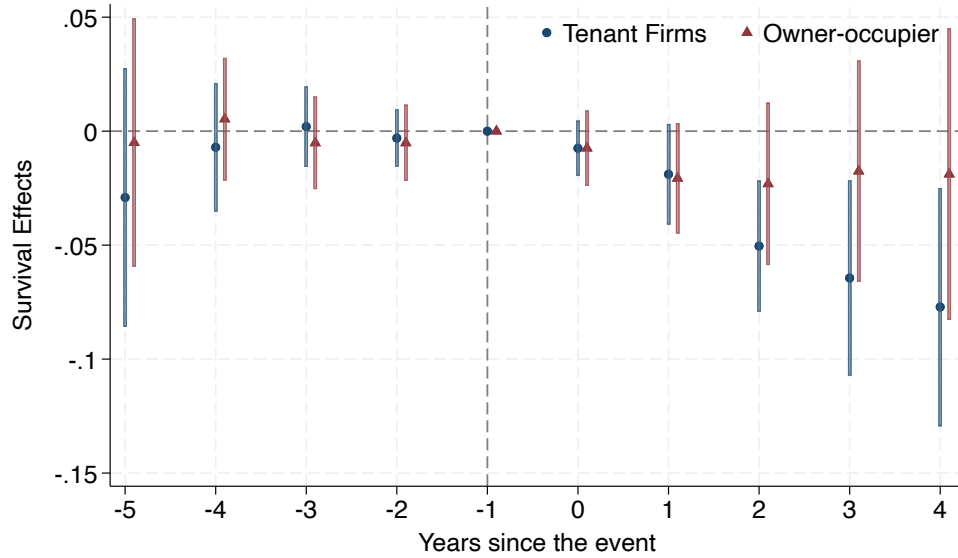


Figure A.17: Survival Effects by Occupancy Status

Notes: The figure presents the effects of the reform on firm survival, separately for tenant and owner-occupier firms. Firms are classified as tenants if they report positive rent expenses in the pre-reform year and as owner-occupiers if they report positive land and building values in their CIT returns. Firms reporting neither are assigned occupancy status based on pre-reform capital intensity, with those below (above) the median classified as tenants (owner-occupiers). The outcome variable is an indicator equal to one if the firm operated at time t , zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars represent the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

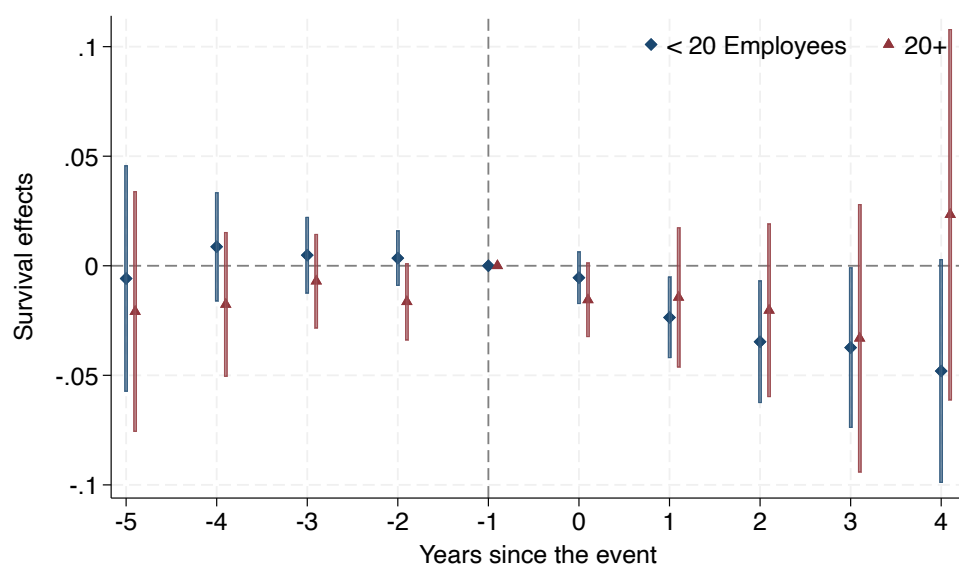


Figure A.18: Survival Effects by Firm Size (Number of Formal Employees)

Notes: The figure presents the effects of the reform on firm survival, separately by firm size, measured using pre-reform formal employment. The outcome variable is an indicator equal to one if the firm operated at time t , zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

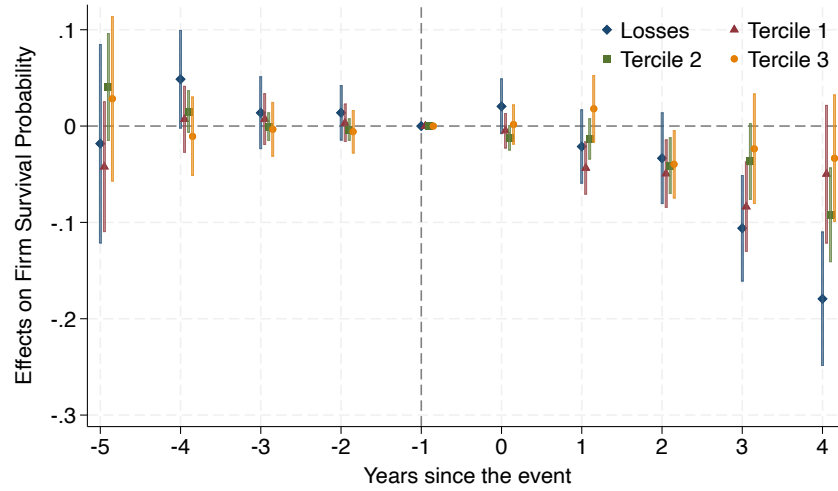


Figure A.19: Survival Effects by Profitability

Notes: The figure presents the effects of the reform on firm survival across different points of the pre-reform profit margin distribution.. The outcome variable is an indicator equal to one if the firm is operating at time t , zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

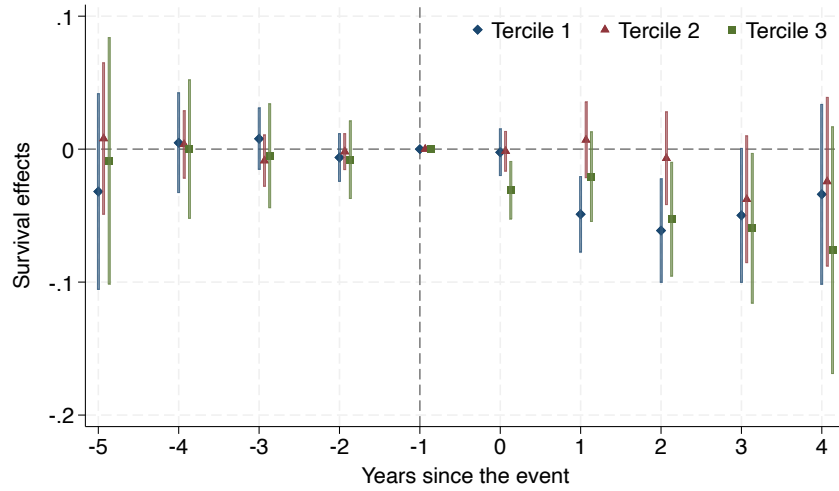


Figure A.20: Survival Effects by Leverage

Notes: The figure presents the effects of the reform on firm survival across firms with different levels of pre-reform leverage (debt-to-asset ratio). The outcome variable is an indicator equal to one if the firm operated at time t , zero otherwise, implying that the coefficients can be interpreted as percentage-point changes in survival probability. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

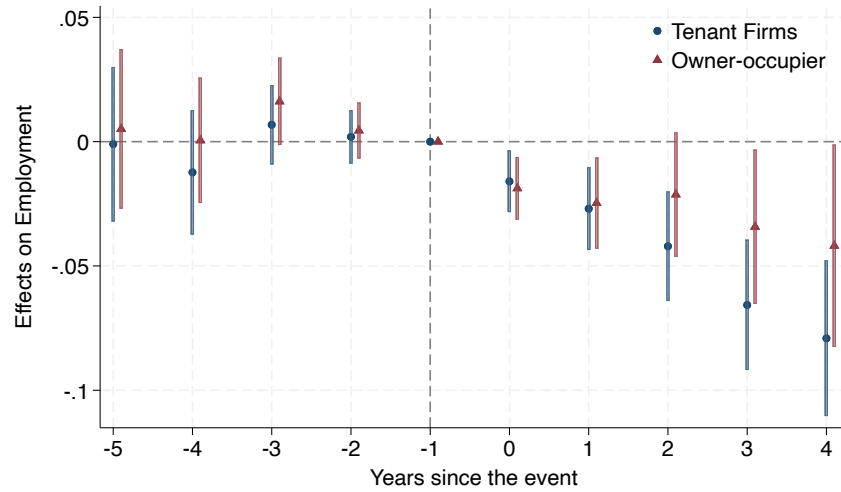


Figure A.21: Employment Effects by Occupancy Status

Notes: The figure presents the dynamic effects of the reform on formal employment, separately for tenant and owner-occupier firms. The outcome variable is the number of workers scaled by its pre-reform value, implying that the coefficients can be interpreted as percentage changes. Firms are classified as tenants if they report positive rent expenses in the pre-reform year and as owner-occupiers if they report positive land and building values in their CIT returns. Firms reporting neither are assigned occupancy status based on pre-reform capital intensity, with those below (above) the median classified as tenants (owner-occupiers). The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using the staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#). Standard errors are clustered at the firm level.

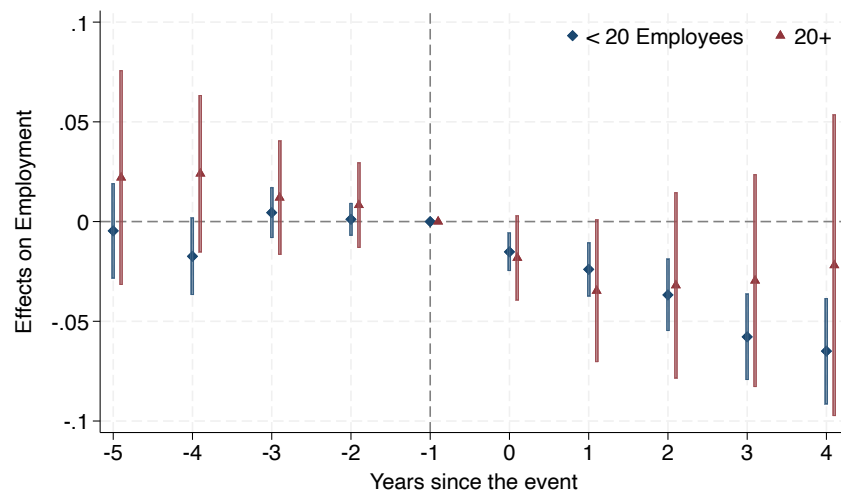


Figure A.22: Employment Effects by Firm Size (Number of Formal Employees)

Notes: The figure presents the dynamic effects of the reform on formal employment, separately by firm size, measured using pre-reform formal employment. The outcome variable is the number of workers scaled by its pre-reform value, implying that the coefficients can be interpreted as percentage changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using the staggered difference-in-differences estimator of [Callaway and Sant'Anna \(2021\)](#). Standard errors are clustered at the firm level.

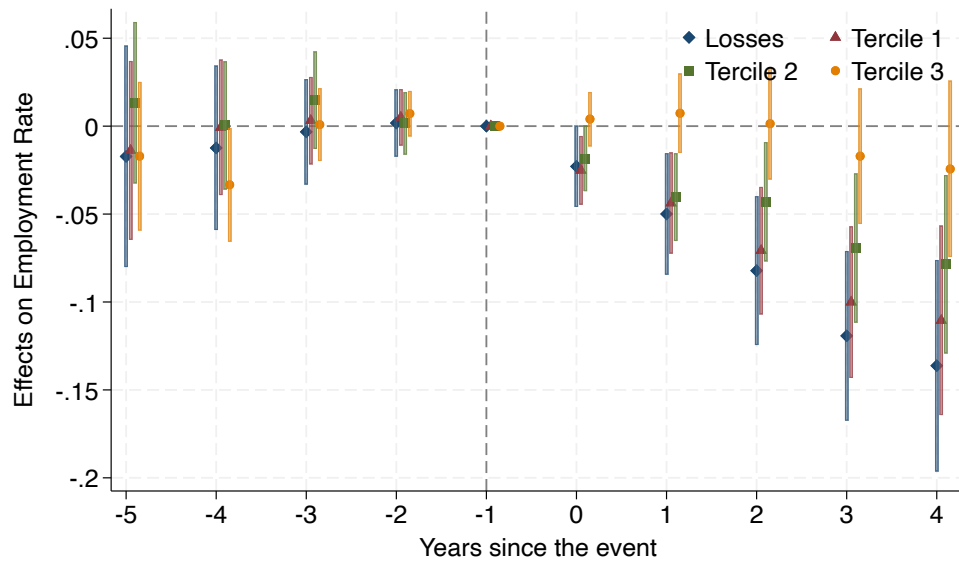


Figure A.23: Employment Effects by Profitability

Notes: The figure presents the dynamic effects of the reform on formal employment across different points of the pre-reform profit margin distribution. The outcome variable is the number of workers scaled by its pre-reform value, implying that the coefficients can be interpreted as percentage changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using the staggered difference-in-differences estimator of [Callaway and Sant'Anna \(2021\)](#). Standard errors are clustered at the firm level.

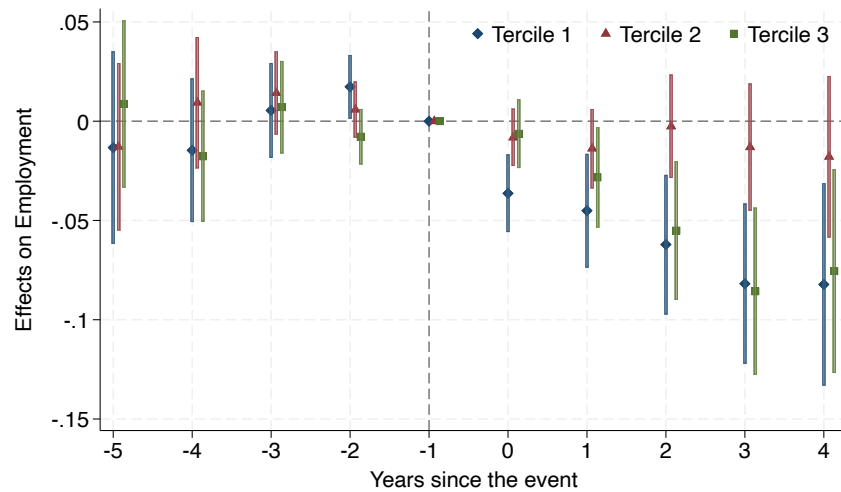


Figure A.24: Employment Effects by Leverage

Notes: The figure presents the dynamic effects of the reform on formal employment across firms with different levels of pre-reform leverage (debt-to-asset ratio). The outcome variable is the number of workers scaled by its pre-reform value, implying that the coefficients can be interpreted as percentage changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1 . The vertical bars indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using the staggered difference-in-differences estimator of [Callaway and Sant'Anna \(2021\)](#). Standard errors are clustered at the firm level.

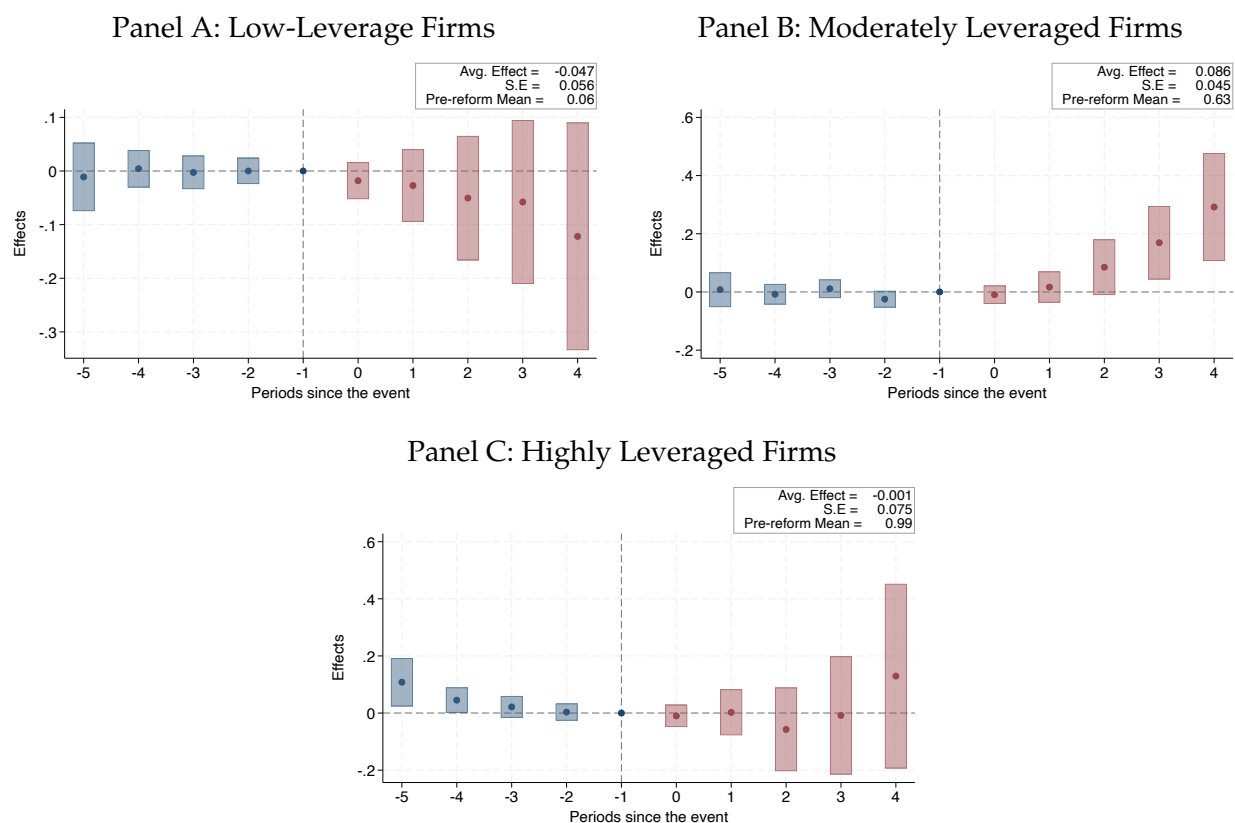


Figure A.25: Effects of the Reform on Firm Debt

Notes: In all panels, the outcome variable is total debt scaled by pre-reform total assets. In Panel A, I restrict the sample to firms in the first tercile of the pre-reform leverage distribution. Panel B restricts the sample to firms in the second tercile, while Panel C restricts the sample to firms in the third tercile (highly leveraged firms). Each regression includes controls for sector of operation, pre-reform firm size (measured by employment), accounting month, and occupancy status (tenant versus owner-occupier). The coefficients are estimated using a staggered difference-in-differences design following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

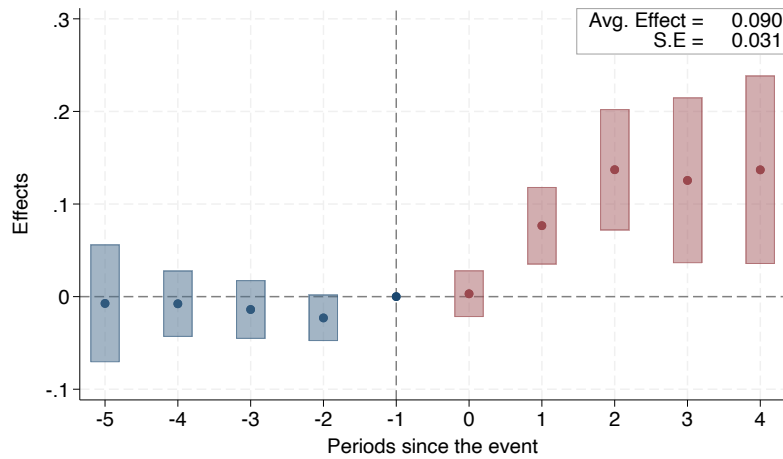


Figure A.26: Effects of Property Taxes on Business Rental Expenses in a Balanced Panel

Notes: The figure presents the dynamic effects of property taxes on firms' rental expenses. The sample consists of a balanced panel of firms that reported positive rental expenses in the pre-reform period. The dependent variable is rental costs scaled by their pre-reform value, implying that the coefficients can be interpreted as percentage changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector, pre-reform size, and accounting month. The coefficients are estimated using a staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Appendix Figure A.27 presents the corresponding event-study estimates using never-treated units as the comparison group. Standard errors are clustered at the firm level.

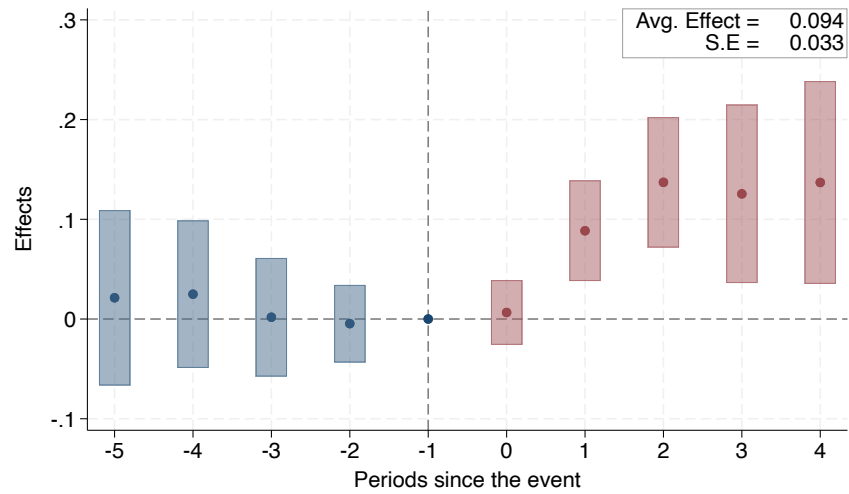


Figure A.27: Effects of Property Taxes on Business Rental Expenses: Never-Treated Firms as the Comparison Group

Notes: The figure presents the dynamic effects of property taxes on firms' rental expenses. The dependent variable is rental costs scaled by their pre-reform value, implying that the coefficients can be interpreted as percentage changes. The horizontal axis indicates the number of years since the reform. Each dot represents the point estimate of the treatment effect for that year, with the reference category set at -1. The vertical bars indicate the corresponding 95 percent confidence intervals. The regression specification includes fixed effects for sector and pre-reform size bins. The coefficients are estimated using a staggered difference-in-differences approach, following [Callaway and Sant'Anna \(2021\)](#), using never-treated units as the comparison group. Standard errors are clustered at the firm level.

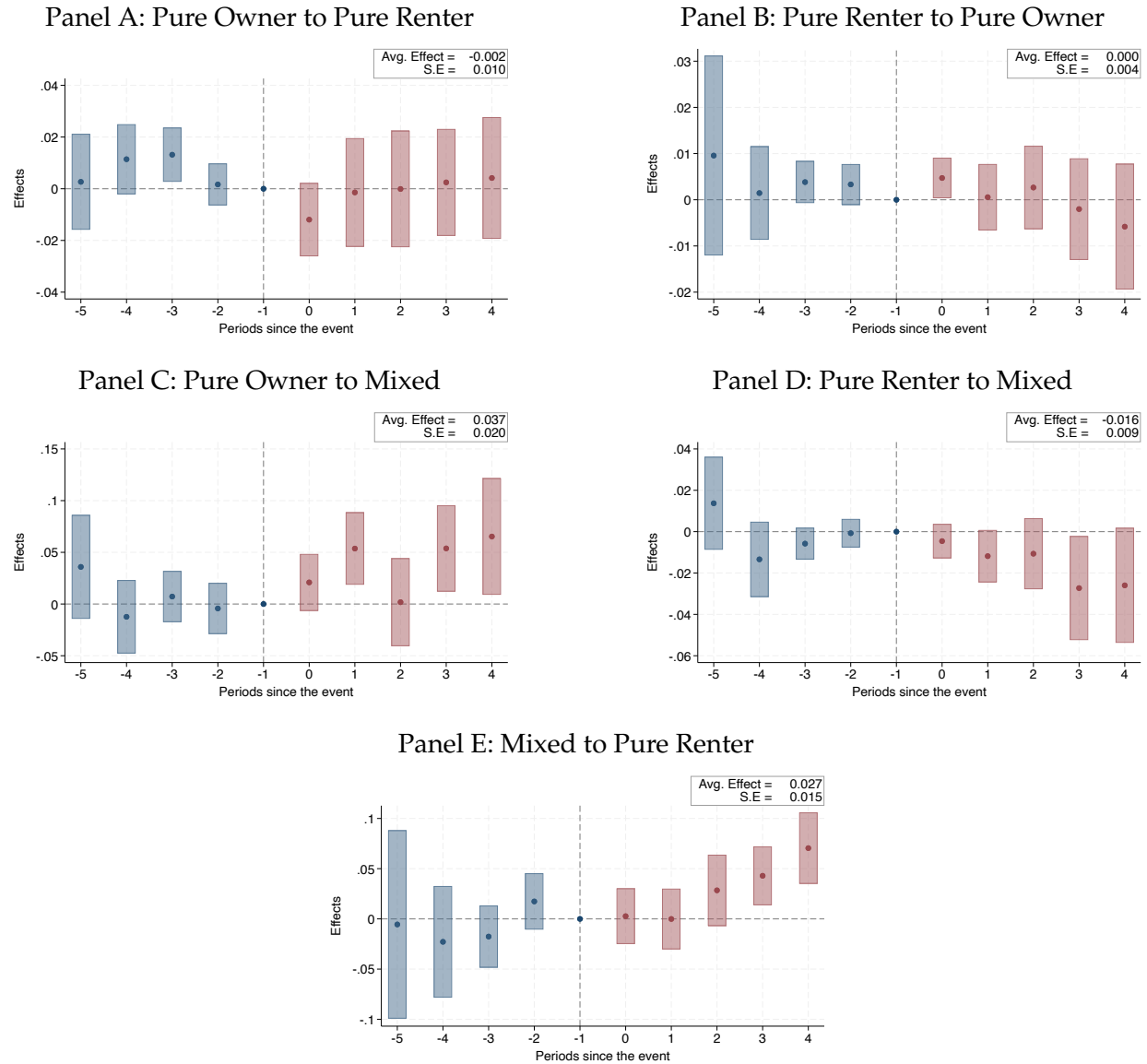


Figure A.28: Effects of the Reform on Occupancy Decisions: *Own or Rent?*

Notes: The figure presents the dynamic effects of the reform on firms' occupancy decisions, examining transitions across occupancy categories at the extensive margin. All outcome variables are binary indicators. The categories are defined as follows: a pure owner is defined as a firm that reports positive land and building values but no rental expenses; a pure renter is a firm that reports positive rental expenses but no land and building values; and a mixed firm reports positive values for both categories. Panel A examines the probability that a pure owner transitions to pure renter following the reform. Panel B examines the probability that a pure renter transitions to pure owner. Panel C examines the probability that a pure owner transitions to the mixed category. Panel D examines the probability that a pure renter transitions to the mixed category. Together, Panels C and D assess whether the reform induced firms to move toward partial occupancy arrangements—simultaneously owning and renting space— from previously exclusive ownership or tenancy arrangements. Panel E examines the probability of a mixed-category firm transitioning to pure renter status. All regressions are estimated using the staggered difference-in-differences estimator of [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Standard errors are clustered at the firm level.

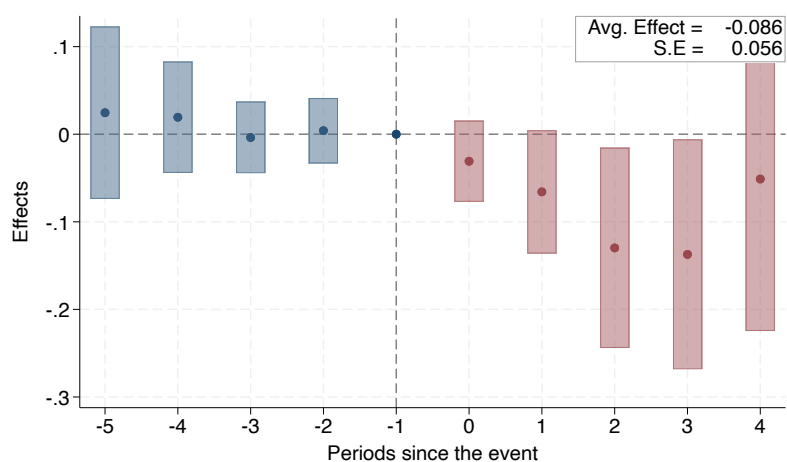


Figure A.29: Reported Land and Building Values: Sample of Pure Owners

Notes: The figure presents the dynamic effects of the reform on reported land and building values among pure owners—defined as firms reporting positive land and building values but no rental expenses in the pre-reform period. The outcome variable is the reported value of land and buildings, scaled by their pre-reform values, such that the coefficients can be interpreted as percentage changes relative to the pre-reform period. The regression specification includes fixed effects for sector and pre-reform accounting month. Standard errors are clustered at the firm level.

I.B. Additional Tables

Table A.1: Effects of the Reform on Firms' Revenues, Profits, Capital, and Investment

	(1) Log Revenues	(2) Profits	(3) Log Capital	(4) Investment/Lagged Capital	(5) Investment: Extensive
ATT	0.013 (0.042)	-0.033* (0.019)	-0.139** (0.065)	-0.027** (0.011)	-0.049** (0.022)
Sector of Operation	Y	Y	Y	Y	Y
Pre-reform Size	Y	Y	Y	Y	Y
Other Controls	Y	Y	Y	Y	Y
Observations	33012	33012	29097	43029	43029

Note: The table reports the average effects of the reform on firms' revenues, profits, capital, and investment. Profits are measured as earnings before interest and taxes (EBIT) (sales minus cost of sales, operating and administrative expenses). EBIT is scaled by pre-reform revenues. Investment is measured as the annual change in fixed assets, scaled by lagged fixed assets. The last column presents the extensive-margin response of investment. The coefficients are estimated using a staggered difference-in-differences approach following [Callaway and Sant'Anna \(2021\)](#), using not-yet-treated units as the comparison group. Each regression includes fixed effects for sector and pre-reform firm size. Under "Other Controls," I include dummies for accounting month and whether the firm filed its returns on time, all measured in the pre-reform period. Standard errors are clustered at the firm level.

Table A.2: Placebo Test: Employment Effects Among Firms Outside Kampala

	(1) Emp/Pre-reform Emp	(2) Log(1+Emp)	(3) IHS(Emp)	(4) Emp in Levels
ATE	0.008 (0.016)	0.026 (0.030)	0.037 (0.037)	-0.065 (0.430)
Industry-Size	Y	Y	Y	Y
Other Controls	Y	Y	Y	Y
Observations	28809	34362	34362	34362

Note: The table reports placebo estimates of the employment effects of the reform. The sample is restricted to firms located outside Kampala City, comparing those within the greater Kampala metropolitan area (GKMA), which are geographically closer to treated firms, to firms located entirely outside the GKMA. All specifications include fixed effects for pre-reform industry-size bins, accounting month, and occupancy status. Standard errors are clustered at the firm level.

Table A.3: Effects on Wage Bills (UGX Millions)

	(1) $\hat{\beta}^{CIT}$	(2) $\hat{\beta}^{PAYE}$	(3) $\frac{ \hat{\beta}^{PAYE} - \hat{\beta}^{CIT} }{ \hat{\beta}^{PAYE} }$
Estimates	-2.290*** (0.872)	-2.660*** (0.777)	0.139
Pre-reform Mean	46.528	43.990	
Observations	108945	108945	

Note: The table reports the effects of the reform on firms' wage bills. Column 1 reports the estimated treatment effect on the total wage bill as reported on firms' corporate income tax (CIT) returns, while Column 2 reports the estimated treatment effect on the formal wage bill constructed from employee tax returns (PAYE). Both coefficients are estimated using the same sample and control variables, with all monetary values expressed in UGX millions. Column 3 reports the share of the decline in the formal wage bill attributable to increased informal wage payments, computed as $\frac{|\hat{\beta}^{PAYE}| - |\hat{\beta}^{CIT}|}{|\hat{\beta}^{PAYE}|}$ using the estimates in Columns 1 and 2. The corresponding event-study estimates are presented in Figure A.12. All specifications include fixed effects for sector, pre-reform size, and accounting month. Standard errors are clustered at the firm level.

Table A.4: Effects on Wage Distribution: 25th, 50th, and 75th Percentiles

	(1) p25	(2) p50	(3) p75
ATE	-0.015 (0.025)	-0.064 (0.045)	-0.184** (0.082)
Sector of Operation	Yes	Yes	Yes
Pre-reform Size	Yes	Yes	Yes
Other Controls	Yes	Yes	Yes
Pre-reform Mean	0.372	0.496	0.816
Observations	19125	19125	19125

Note: The table reports the average effects of the reform on within-firm wage quantiles. The dependent variables are the 25th, 50th, and 75th percentiles of the within-firm wage distribution, scaled by the firm's pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i 's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i 's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the firm's pre-reform wage bill. The sample is restricted to firms with at least five workers in the pre-reform period. All models include fixed effects for sector and pre-reform firm size. Under "Other Controls," I include dummies for accounting month, whether the firm reported positive land and building values, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

Table A.5: Effects on Wage Distribution: 20th, 40th, 60th, and 80th Percentiles

	(1) p20	(2) p40	(3) P60	(4) p80
ATE	-0.027 (0.026)	-0.034 (0.033)	-0.110* (0.064)	-0.197** (0.099)
Sector of Operation	Yes	Yes	Yes	Yes
Pre-reform Size	Yes	Yes	Yes	Yes
Other Controls	Yes	Yes	Yes	Yes
Pre-reform Mean	0.351	0.439	0.583	0.950
Observations	19125	19125	19125	19125

Note: The table reports the average effects of the reform across the within-firm wage distribution. The dependent variables are the 20th, 40th, 60th, and 80th percentiles of the within-firm wage distribution, scaled by the firm's pre-reform formal wage bill, i.e., $\frac{q_{i,t}}{W_{i,0}} \times 100$, where $q_{i,t}$ denotes firm i 's q^{th} wage quantile at time t and $W_{i,0}$ denotes firm i 's pre-reform wage bill. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to the firm's pre-reform wage bill. The sample is restricted to firms with at least five workers in the pre-reform period. All models include fixed effects for sector and pre-reform firm size. Under "Other Controls," I include dummies for accounting month, whether the firm reported positive land and building values, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using the staggered difference-in-differences estimator of Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

Table A.6: Property Taxes and Worker Composition

	(1) < p25	(2) p25-50	(3) p50-75	(4) p75-100
Average Effect (in pp)	0.404 (1.213)	0.698 (1.046)	2.212** (0.879)	-2.838** (1.394)
Sector of Operation	Y	Y	Y	Y
Pre-reform Size	Y	Y	Y	Y
Other Controls	Y	Y	Y	Y
Pre-reform Mean	21.411	22.885	24.831	29.341
Observations	19125	19125	19125	19125

Note: The table reports the average effects of the reform on within-firm employment composition. The dependent variables are the shares of employment in each quartile, relative to pre-reform total employment, i.e., $\frac{E_{i,q,t}}{E_{i,0}} \times 100$, where $E_{i,q,t}$ denotes firm i 's number of workers in quartile q at time t and $E_{i,0}$ denotes firm i 's total formal employment in the pre-reform period. Accordingly, the coefficient estimates should be interpreted as percentage-point changes relative to firms' pre-reform total employment. The sample is restricted to active firms with at least five workers in the pre-reform period. All models include fixed effects for sector and pre-reform firm size. Under "Other Controls," I include dummies for accounting month, whether the firm reported positive land and building values, and whether the firm filed returns on time, all measured in the pre-reform period. The coefficients are estimated using the staggered difference-in-differences approach following Callaway and Sant'Anna (2021). Standard errors are clustered at the firm level.

Table A.7: Implications for Central Government Tax Revenue

	CIT		Employee Tax		Combined	
	(1)	(2)	(3)	(4)	(5)	(6)
ATE	-0.085 (0.109)	-0.091 (0.129)	-0.543*** (0.126)	-0.451*** (0.141)	-0.627*** (0.188)	-0.542** (0.217)
Controls	No	Yes	No	Yes	No	Yes
Pre-reform Mean	1.455	1.455	6.355	6.355	7.810	7.810
Observations	137862	137862	137862	137862	137862	137862
Central Revenue Loss per UGX 1					0.295	0.255

Note: The table reports the average effects of the reform on central government tax revenue. All analyses are aggregated at the firm level. The outcome variables are measured in millions of constant 2016/17 Uganda shillings (UGX). Columns 1 and 2 report the effects on corporate income tax revenue, Columns 3 and 4 report the effects on employee tax revenue (PAYE), while Columns 5 and 6 report the effects on combined tax revenue. The sample is a panel of all firms that were active in the pre-reform year, with tax revenue set to zero following firm exit. The regressions control for sector, accounting month, and occupancy status (tenant versus owner-occupier). Standard errors are clustered at the firm level.

II. A FRAMEWORK FOR PROPERTY TAX PASS-THROUGH

The finding that higher property tax liabilities increase business rental expenses contrasts with the predictions of neoclassical theory, which suggests little to no pass-through of property taxes ([Mieszkowski 1972](#); [Zodrow 2001](#)). Underlying the neoclassical prediction is the assumption that demand for commercial space within a jurisdiction is elastic, since other jurisdictions offer comparable commercial units, while supply is inelastic. In that setting, the property tax falls on the immobile factor and landlords absorb the entire burden because attempts to raise rents would induce tenants to relocate to competing jurisdictions.

A necessary condition for this result, however, is that the property tax is a cost of *owning* the property, not a cost of *renting it out*. If the tax is enforced universally, the landlord owes the same liability whether the property is occupied or vacant, and the tax is effectively a lump sum that cannot influence the landlord's pricing decision. But if the tax is enforced selectively—collected only when the property is occupied and generating observable economic activity—then renting out the property triggers the tax liability. In this case, the tax functions as a marginal cost of supplying space to the rental market, and standard incidence logic predicts positive pass-through to tenants.

To rationalize my findings, I develop a simple framework that predicts non-zero pass-through and that does not depend on the elasticities of demand and supply in the standard sense. The framework embeds a key institutional feature of property tax enforcement in developing countries: *selective enforcement*, where enforcement and collection probabilities are lower for vacant properties than for occupied properties. This pattern arises naturally in settings where tax administrations have limited capacity and prioritize enforcement on properties that are visibly in use and generating income ([Franzsen and McCluskey 2017](#)). Under limited administrative capacity, targeting occupied properties becomes a cost-effective enforcement strategy: occupied properties are easier to identify, their owners are easier to locate, and the revenue yield per enforcement action is higher because an active tenant signals ability to pay. Consistent with the pattern of selective enforcement documented above, [Ahabwe et al. \(2020\)](#) notes that demand letters in Kampala are disproportionately directed toward high-value property owners, reflecting the administrative reality that limited enforcement resources incentivize tax authorities to prioritize collections where the potential yield is greatest.

Setup:— Consider a landlord who owns a unit of commercial space and must choose the rent R to charge. If the unit is occupied at rent R , the landlord earns rental income R

and pays the property tax liability $T = \tau V$ with certainty, yielding a payoff of $R - T$.²¹ If the unit remains vacant, the landlord earns no rental income but may still face the tax. The tax is enforced on the vacant unit with probability $p_e \leq 1$, yielding an expected payoff of $-p_e T$.²²

The probability that the unit is occupied depends on the rental price charged. Let $q(R)$ denote the probability that the unit is filled at rent R , with $q'(R) < 0$ (higher rents reduce the likelihood of finding a tenant) and $q(0) \leq 1$. Assuming a unit mass of potential tenant firms, $q(R)$ also represents the aggregate demand for commercial space, i.e., the share of businesses willing to pay rent R rather than take their outside option.²³

Landlord's Maximization Problem:— The landlord sets rent to maximize his expected utility:

$$\max_R q(R) (R - T) + (1 - q(R)) (-p_e T) \quad (4)$$

The first term is the expected payoff if the unit is filled: the landlord earns $R - T$ with probability $q(R)$. The second term is the expected payoff if the unit remains vacant: the landlord pays $p_e T$ in expected tax with probability $1 - q(R)$.

Expanding (4) and dropping the terms that do not depend on R , the landlord effectively maximizes:

$$\max_R q(R) [R - (1 - p_e)T] \quad (5)$$

This formulation has a straightforward economic interpretation: the landlord behaves as if facing a per-unit cost of $(1 - p_e)T$ for each occupied unit. This cost is the additional expected tax the landlord incurs by having the unit occupied rather than vacant. Under universal enforcement ($p_e = 1$), this cost is zero, i.e., the landlord pays T regardless of occupancy, so occupancy triggers no additional liability. Under fully selective enforcement ($p_e = 0$), the cost is the full tax T , i.e., each occupied unit triggers the entire tax bill.

The first-order condition of (5) is:

$$q'(R) [R - (1 - p_e)T] + q(R) = 0 \quad (6)$$

²¹ τ is the property tax rate and V is the assessed value.

²²In a setting with imperfect compliance, even when the unit is occupied, as is common in many developing countries, p_e can be interpreted as the relative probability of compliance when the unit is vacant compared to when it is occupied. If we assume landlords are more likely to comply when the unit is occupied than when it is vacant, then $p_e < 1$.

²³This equivalence holds when each potential tenant independently decides whether to rent the unit. If tenants have heterogeneous reservation values for the location, $q(R)$ is the fraction whose reservation value exceeds R .

Solving for R :

$$\boxed{R^* = (1 - p_e)T - \frac{q(R^*)}{q'(R^*)}} \quad (7)$$

The optimal rent has two components. The first, $(1 - p_e)T$, is the effective marginal cost of occupancy—the portion of the tax that the landlord can avoid by leaving the unit vacant. The second, $-q(R^*)/q'(R^*)$, is the standard markup that depends on the shape of the demand curve. Under universal enforcement ($p_e = 1$), the first term vanishes and the optimal rent depends only on demand conditions, not on the tax, recovering the neoclassical result.

Characterizing the Pass-through Rate:—Defining $c \equiv (1 - p_e)T$ as the effective marginal cost of occupancy, the landlord's first-order condition implicitly defines the optimal rent R^* as a function of the effective cost c , and hence of T . Define the function:

$$G(R, c) \equiv q'(R)(R - c) + q(R) = 0 \quad (8)$$

By the implicit function theorem:

$$\frac{dR^*}{dc} = -\frac{\partial G/\partial c}{\partial G/\partial R} \quad (9)$$

Computing the partial derivatives:

$$\frac{\partial G}{\partial c} = -q'(R)$$

$$\frac{\partial G}{\partial R} = q''(R)(R - c) + q'(R) + q'(R) = q''(R)(R - c) + 2q'(R)$$

Therefore:

$$\frac{dR^*}{dc} = -\frac{-q'(R^*)}{q''(R^*)(R^* - c) + 2q'(R^*)} = \frac{q'(R^*)}{q''(R^*)(R^* - c) + 2q'(R^*)} \quad (10)$$

From the first-order condition (6), we can substitute $R^* - c = -q(R^*)/q'(R^*)$:

$$\frac{dR^*}{dc} = \frac{q'}{q'' \left(-\frac{q}{q'} \right) + 2q'} = \frac{q'}{-\frac{q q''}{q'} + 2q'} = \frac{(q')^2}{-(q)(q'') + 2(q')^2} \quad (11)$$

Dividing numerator and denominator by $(q')^2$:

$$\frac{dR^*}{dc} = \frac{1}{2 - \frac{q(R^*)q''(R^*)}{[q'(R^*)]^2}} \quad (12)$$

Since $c = (1 - p_e)T$ and hence $dc/dT = (1 - p_e)$, the pass-through rate with respect to the tax liability is:

$$\rho = \frac{dR^*}{dT} = \frac{dR^*}{dc} \cdot \frac{dc}{dT} = \frac{1 - p_e}{2 - \theta}, \quad (13)$$

where $\theta \equiv \frac{q(R^*)q''(R^*)}{[q'(R^*)]^2}$ is the demand curvature parameter introduced in the tax incidence literature by [Weyl and Fabinger \(2013\)](#).

The parameter θ governs the distribution of the tax burden for a given $p_e \in [0, 1]$. The pass-through rate is increasing in θ for any given p_e . Equation 13 also permits over-shifting if $\theta > 1$ (strongly log-convex demand) and p_e is sufficiently small. If we assume a simple, linear demand, $q(R) = q_0 - q_1 R$, we have $q'' = 0$ and therefore $\theta = 0$. The pass-through rate is $\rho = (1 - p_e)/2$. Under this special case, the burden of property taxes is equally shared between the landlord and the tenant if $p_e = 0$, the case in which there is no enforcement on vacant units. Under linear demand and strictly no enforcement on vacant properties, the landlord bears more than half the burden when $\theta < 0$ (log-concave demand), while the tenant bears more than half when $\theta > 0$ (log-convex demand).

Note that the pass-through formula (13) requires $\theta < 2$ for $\rho \geq 0$. This restriction follows from the second-order condition for the landlord's optimization problem, which ensures that the optimal rent R^* is a well-defined interior maximum. To see this, note that the second-order condition requires:

$$\frac{d^2}{dR^2} [q(R)(R - c)] = q''(R)(R - c) + 2q'(R) < 0 \quad (14)$$

From the first-order condition (6), we can substitute $R^* - c = -q(R^*)/q'(R^*)$:

$$q''(R^*) \left(-\frac{q(R^*)}{q'(R^*)} \right) + 2q'(R^*) = -\frac{q(R^*)q''(R^*)}{q'(R^*)} + 2q'(R^*) \quad (15)$$

Factoring out $q'(R^*)$:

$$= q'(R^*) \left(2 - \frac{q(R^*)q''(R^*)}{[q'(R^*)]^2} \right) = q'(R^*) (2 - \theta) \quad (16)$$

Since $q'(R^*) < 0$, the expression in (16) is negative if and only if $2 - \theta > 0$, i.e., $\theta < 2$. When $\theta \geq 2$, the landlord's objective function is locally convex at the candidate solution, and no interior optimum exists.